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Bidding Bazaar

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Abstract

In today's rapidly evolving digital landscape, the car auction process has been characterized as inefficient and cumbersome. Recognizing the pressing need for modernization in this area, we proudly introduce an innovative online bidding platform. Our platform, driven by Artificial Intelligence (AI) technology, plays a pivotal role in predicting car prices, making the auction process simple and efficient for users. By harnessing the analytical prowess of AI, we aim to create a marketplace that surpasses the limitations of traditional methods, offering enhanced efficiency and a positive user experience.

Executive Summary

Bidding Bazaar is aimed at addressing the issues faced by traditional and existing platforms and providing a better approach toward buying and selling cars. Firstly, traditional methods of buying and selling cars are often inefficient and inconvenient. Buyers and sellers have to physically visit dealerships multiple times to meet multiple sellers or buyers. This traditional approach is not convenient and efficient and costs a lot of time for both buyer and seller. Secondly, the existing car trading platforms and traditional methods lack transparency. Buyers may not have access to accurate pricing information while sellers may not be aware of the current market value of their car. Price determination can be a major hurdle. Not knowing the fair price often creates misunderstanding. There are some tools available online that may provide prices for cars but they don't consider the current condition of cars thus they provide inaccurate information. In traditional methods the market reach is limited. Sellers may not find a fair offer for their car with limited reach. Similarly, buyers may not find the car they are looking for or not the condition they are looking for in limited reach.

To solve these issues Bidding Bazaar is aimed at streamlining the car trading process, enhancing user experience, and improving transparency. Additionally, the platform will have an integrated AI-based predictor to predict current market value added to facilitate users. Our platform addresses the issue of physical multiple visits by providing them with a digital platform where sellers can create listings of their cars and buyers can explore and place bids on relevant cars. The availability of inspection reports with each car eliminates the need to visit physically. To solve the issue of transparency and price determination, Bidding Bazaar uses an AI-based price predictor to predict the market value of cars based on the current conditions of the car. The verification of car listing requests by the admin before they are listed on the platform reduces the chances of future misunderstandings between sellers and buyers.

Bidding Bazaar is designed in a way that is open to continuous improvement and integration of new features based on user feedback. This ensures that the platform will remain effective and competitive in the future car market.

In conclusion, "Bidding Bazaar" is designed and aimed to address the issues of traditional and existing methods. Bidding Bazaar resolves these issues by streamlining the trading process, enhancing transparency through AI-based pricing, and enhancing user experience. The project aims to redefine the car trading landscape and make an efficient and transparent platform for both buyers and sellers.

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Chapter 1 Introduction

The traditional methods for buying cars have historically relied on visiting dealerships and negotiations. This made the process time-consuming and difficult. Online platforms sought to address these issues by introducing a digital platform, but buyers still had to visit and check the car and transparency remained a problem.

To overcome these issues our project is dedicated to redefining the car trading experience. By providing a Bidding platform with integrated AI-based price prediction, we aim to streamline the process, enhance user experience, and provide transparency by providing fair market value for vehicles.

1.1 Purpose of this Document

This document provides a detailed technical and general overview of our FYP, Bidding Bazaar. This document will provide details about every corner of our project, including analysis, objectives, design, implementation, and testing as well as limitations of our project. For users to sell and buy their cars at the best prices, this project aims to build an MVP with an AI-powered car price predictor. The application aims to assist users with the predicting price of the vehicle and provide them with a user-friendly interface and seamless user experience when buying and selling their cars through our AI-powered platform.

1.2 Intended Audience

The intended audience for this document are academic audience which is mainly composed of a panel of professors or educationalists looking to understand and evaluate this project. In addition to this, any concerned person who wants to comprehend the system's operations can review certain parts of this report. For the general non-technical audience seeking to use the platform, a simpler version will be released which will somewhat act as a user guide.

1.3 Definitions, Acronyms, and Abbreviations

Important definitions, acronyms, and abbreviations used in this document.

SDG: Sustainable Development Goal

FYP: Final Year Project

MVP: Minimum Viable Product

AI: Artificial Intelligence

UX: User Experience

CRUD: Create, Read, Update, Delete

1.4 Conclusion

In conclusion, from the standpoint of our project for buying and selling cars traditional methods are not adequate in the rapidly changing modern era. While there are some online platforms available, they are still inefficient and lack transparency. We aim to develop a system that addresses these issues. The end users as a seller can create listings and potential buyers can bid on listings. This eliminates the need to physically visit each car, thus reducing the time required for selling and buying and streamlining this process. Other than that, the price prediction feature of our system will significantly increase price transparency, benefiting both seller and buyer in the estimation of the current market value of the car.

Chapter 2 Project Vision

This chapter will provide an overview of the project problem domain, the project statement and its elaboration, the goals and objectives, and the scope of the project.

2.1 Problem Domain Overview

The problem domain of this project revolves around creating a user-friendly and seamless platform that helps both sellers and buyers with AI-based price prediction.

The system first requires the user to sign up and register. As a seller, the user can request to list the car by providing details of the car and scanned copies of all the original documents. After verification of details by admin, a physical inspection request is initiated. After inspection, the car is rated based on the current condition of various components. These ratings provide information about the current condition of the car to the potential buyer and help in better predictions of car prices by the system. After this seller has to enter a price comparable to the price predicted by the system and the car is listed in the system. The system starts bidding after the listing is made and ends at a given time. As a buyer, the system shows all the active listings. During the active period, the system allows potential buyers to place bids on the car. The System notifies both the seller and the buyer after the end time. After bidding, the seller and buyer then have to meet to make payment and receive the car. Post-transaction buyer and seller can rate each other on the system.

The ultimate goal is to provide users with a user-friendly website with AI-based price predictor to enhance user experience and to help both buyer and seller determine the near actual value of the vehicle. It will enhance user experience, streamline the vehicle selling and buying process, and establish a platform built on trust and efficiency.

2.2 Problem Statement

The existing methods and platforms for car selling and buying are inefficient and lack transparency. To address these issues, we are developing a car bidding system that will streamline the process and increase transparency. To improve user experience even further by assisting sellers and buyers in determining the price of cars, we integrated AI-based price prediction into our system.

2.3 Problem Elaboration

According to our research, the existing methods and platforms for car buying and selling cars face numerous challenges. These challenges include inefficiency and transparency. Firstly, the inefficiency of traditional methods and existing platforms can be attributed to the need for physical visits to check every car, even if the car is physically inspected, there can still be price negotiation issues. Secondly, another issue is the lack of transparency. Buyers often struggle to estimate the fair price of a vehicle. On the other hand, sellers may find it difficult to estimate the right market price for their vehicle.
According to our research, there is not a single platform that solves these issues, so we aim to develop a bidding system that solves these problems. It solves inefficiency by providing a platform for sellers and buyers of verified and inspected cars and transparency by integrating an AI-based price prediction system to help both seller and buyer determine the fair price of the vehicle.
Our system is going to target the following issues that currently prevail in the auctioning of cars.

2.3.1 Inefficient Process

Currently, the process of auctioning a car is inefficient and cumbersome. There is a lot of manual work involved: To put your car for auction you have to get it inspected, drive it to the auction site, and it is kept there until it's purchased or the auction has ended.
Our system makes all of this process a lot easier by making the auction online. The user sitting in their home can post a car for auction. The buyers can bid on it online. They can engage in conversations online through our chat feature. If someone wants to sell their cars more quickly then they can buy the premium features where their car gets features and is pushed to more relevant buyers.

2.3.2 Non-Serious Buyers and Sellers

Our system is for serious buyers. In existing platforms, people put up fake listings. Bidding Bazaar aims to solve this issue by first verification of listings by admin.

2.3.3 Car's Pricing

In the process of auctioning and selling a car online, the most time-consuming and frustrating thing is to set a base price for your car. For sellers, it is often frustrating and time-consuming to set up a price for their car. Also, buyers are uncertain about the price at which they should buy a car. Our system resolves this issue by introducing an AI-based prediction feature, which gives an accurate base price for the car. This makes the process easier and faster for users and they can put their car up in seconds. By putting accurate prices more cars get sold and bought, which is a win-win for everyone.

2.4 Goals and Objectives

The main goal of our platform is to make the process of buying and selling cars efficient and smooth. We want sellers to put accurate prices on their cars and buyers to be confident when purchasing.

- Enhance car auction efficiency through streamlined processes.
- Develop a user-friendly web app for accessible and secure auctions.
- Use AI to help the user buy and sell cars.
- Revolutionize the auction industry through innovation and efficiency.
- Solve the problem of fake listings to an extent.

2.5 Project Scope

We will be creating a full end-to-end website, where a seller can put up a car to sell, get a price with the help of AI, and the buyer can bid on it. The system includes three entities the Buyer, Seller, and Admin with their different user roles and features.

The system will allow users to :

- Post a car on the platform to bid.
- Allow setting a base price for their car.
- Bid on posted cars.
- Have a user-friendly and seamless experience

The price prediction algorithm will be implemented in Python and deployed on a different server. The predicted result will be fetched from this server to our main website. The main website will be in JavaScript, with the frontend built using React JS, the backend built in Express and Node JS, and the Database in MongoDB. For project management, we will use Notion.
At the end of project we will deliver:

- A website for people who want to buy or sell cars.
- A user manual for the application

The project scope does not include the following:

- Providing hosting or server infrastructure for the application
- Developing mobile application (iOS or Android)

2.6 Sustainable Development Goal (SDG)

Our sustainable development goal is Industry, Infrastructure, and Innovation. We want to revolutionize the car auction industry through our innovation. We aim to use technologies like Artificial Intelligence to bring innovation to our product.



Figure 2.1: Industry, Innovation and Infrastructure

The figure represents SGD-9

2.7 Constraints

There are a few constraints that we would be following, mentioned below :

- Time: The project must be completed within a time limit of two semesters (12 months).
- Work Resources: The project will be developed by a group of three members.
- Scope: The scope of the project should be according to the time limit of 12 months.
- Quality: The website should work properly, and be user-friendly.

2.8 Business Opportunity

This product can become a business in itself. The business model could be built by charging users a premium for some premium features like AI-based prices, featuring their cars. Moreover, this product can also be sold as software to large auction companies that want to automate their processes and use AI to decide the auction prices of cars. Large auction companies that sell thousands of cars a day can benefit from this product. It will help them save time and help them set an accurate price, which will

eventually improve their revenue.

2.9 Stakeholders Description/ User Characteristics

The stakeholders for the system would be the Customers (buyers and sellers) and the Admin.

2.9.1 Stakeholders Summary

The customers of this product are the people who want to sell or buy cars through our system. They will have their panel, where they can view cars posted by other sellers and post their cars. The customers can view posted cars and bid on them. Whoever bids the highest at the end of a certain time limit will win the bid

The admin will have his panel with extra control. The admin can create new admins and approve bids

2.9.2 Key High-Level Goals and Problems of Stakeholders

2.9.2.1 Efficiency

This system will help users to efficiently sell and buy cars. It saves their time as they don't need to worry about the price they set while selling or the price they buy a car at. The system takes care of it so that all the users are satisfied.

2.9.2.2 Automating Process

This system automates the whole process of auctioning a car. All the user has to do is post the car on our platform. There is minimal user interaction as we have a chat feature, where sellers and buyers can engage.

2.9.2.3 Predict Accurate Car Prices

This system will help users to set up an accurate price for their car. When the user posts a car the system will generate a base price.

2.10 Conclusion

In conclusion, the vision of our project is to make the process of auctioning cars efficient and user-friendly. We want to revolutionize this industry with the use of AI. This project aims to help our cus-

tomers sell and buy cars in a matter of seconds. Our project scope is to deliver a functional website that includes all the features already addressed.

Chapter 3 Literature Review / Related Work

The following section contains details about the related work done. It has information about some of the previously built systems which somehow relate to Bidding Bazaar. Our primary objective to include this section is to look at the previous software. It will the readers to get an idea and knowledge of the current similar systems or websites like Bidding Bazaar. Knowledge of the previous work can significantly improve our system.

3.1 Definitions, Acronyms, and Abbreviations

Some of the important acronyms, abbreviations, and definitions are given below:
AI: Artificial Intelligence
SEO: Search engine optimization
MAE: Mean Absolute Error
LightGBM: Light gradient-boosting machine
etc: Et cetera
MLR: Multiple linear regression
API: Applied programming interface
R-Squared: Coefficient of determination
Web App: Web-based application

3.2 Price Prediction

In this section, we will go through some of the existing price-prediction models. We will also see how they operate.

3.2.1 Price prediction of used cars using machine learning

The research paper under consideration [1] clearly shows that the prices of the used or old cars can be forecasted by AI models to an acceptable accuracy. Their goal was to predict the price with the highest level of accuracy. For this purpose, they implemented several different prediction models. The conclusion of the study states that among all the models which authors used random forest gave the best results. Its accuracy was determined through the R-square value. It was much better than other models as its at 0.90. Prediction models take many different parameters to predict the price. However, accuracy also depends upon the type of the dataset.

3.2.2 Price prediction of used cars based on LightGBM

The proposed paper [2] predicts car prices with the help of LightGBM. It uses transaction data from an old car-selling platform. In this paper, researchers have used the LightGBM model to predict the price. Grid search helped in improving the accuracy of this model. Results suggest LightGBM is more accurate than classical regression algorithms. This approach will help in reducing transaction risks. It will also improve this niche segment of the market.

3.2.3 Price prediction using regression models

The study in [3] shows different models have different performances. The researchers conducted a comparative study in this regard. The data was scraped from electronic commerce stores. These electronic commerce stores were mostly from Germany. The supervised machine learning models were the main target of this research. Especially the various regression models such as RF regression, MLR, and gbrt. The best performance was of gbrt. Its gave the MAE of nearly 0.3. The MAE of MLR was 0.55. The MAE of RF regression was 0.35.

3.3 Online Car Marketplaces

In this section, we will look at some of the online car marketplaces. There are many such websites, but only a few which are helpful for the users.

3.3.1 Pakwheels

Pakwheels [4] is one of the leading virtual car marketplaces. It has millions of active users. On average, there are eighty thousand active listings on this platform. Their goal is to revolutionize how people buy and sell cars. Pakwheels has integrated many powerful tools to help its users. The thing that makes Pakwheels different from its competitors is its continuous innovation. The only thing that Pakwheels is missing is a bidding system. A bidding system can boost the revenue of their site. Users can not hide their phone numbers from others, which is another problem of Pakwheels. Since anyone can call a particular user, it can cause problems. It can cause disturbance for females and people with busy schedules. Our platform aims to solve these problems for our users.

3.3.2 eBay Motors

eBay [5] has been around for quite some time. It has millions of daily users. It has a feature through which interested buyers can make an offer to the seller. A person without any previous knowledge will find it overwhelming as their guidance regarding pricing. eBay has some great features, but it mainly

targets developed countries. It currently does not have any intentions to expand its operations into underdeveloped countries. To make the most of this website, it's essential to have a good understanding of market prices in order to secure a more advantageous deal. It gives more facilities to buyers than sellers. The absence of a mechanism to ascertain the car's base price, coupled with the seller's hidden base price, complicates the decision-making process.

3.3.3 AAAJapan

AAAJapan [6] stands for all auto auction Japan. AAAJapan is a car auction platform of Japan. Users can place bids on old cars. But it has an old interface. Plus, it does not guide users regarding the price of a particular car. For a new user, it can be overwhelming. To use this site users should accurately know what they are looking for. It takes time to understand this platform. Our system will be easy to use with a far better graphical interface. Our system will also help the user in making their decisions. I will assist by utilizing AI to forecast the cost.

3.3.4 Olx

Olx [7] is an application where people can list their old belongings. Olx offers an old car section, but it lacks tools and features for its users. Anyone can see the phone number of the car owner. There are a lot of fake listings on Olx. Our platform will be moderated so that there are no fake listings. Currently, Olx only moderates the removal of outdated listings and does not attend to other content. Bidding Bazaar will be AI-powered to assist users. Furthermore, our platform will save time for busy individuals, eliminating the necessity to answer every call and engage in negotiations. Anyone will be able to sell their car in a safe online environment.

3.3.5 Gari.pk

Gari. pk [8] is a Pakistani platform where people can list their cars. It lacks a bidding system and contains numerous advertisements, significantly impacting the user experience. Additionally, it lacks a rating system for buyers or sellers and has a less user-friendly interface, making it challenging to use. On Gari. pk, the only way to make an offer is by contacting the listing owner through a phone call. Gari.pk encounters challenges in establishing sustainable revenue models, as the bulk of its income is derived from Google ads displayed on its website. In contrast, our platform will employ a stable business model, eliminating the need for online ads.

3.3.6 Sehgal Motorsports

Sehgal Motors [9] is one of the most followed automobile channels in Pakistan. They have a website for their business. In addition to this a bidding website is also registered under their name. In reality their bidding site is just a static page. Google has positioned it at the top due to its substantial following and robust SEO efforts. In addition to this, they have no tools for their users. Their bidding website is neither user-friendly nor up-to-date.

3.4 Related Work Summary Table

Bidding Bazaar will be an up-to-date, modern, AI-powered, and user-friendly application. Table 3.1 shows a comparison between different car-selling applications. The latest web application should have

Table 3.1: Relative Competitors

Comparison of different car selling platforms

Name	Bidding System	Up to Date Website	AI Powered	User Friendly	Listing moderation
Sehgal Motorsports [9]	No	No	No	No	No
eBay Motors [5]	Yes	Yes	No	Yes	No
Pakwheels [4]	No	Yes	No	Yes	No
Olx [7]	No	Yes	No	Yes	No
Gari.pk [8]	No	No	No	No	No
AAAJapan.com [6]	Yes	Yes	No	No	Yes

all the above features. In addition to this the given below table shows the comparison between different datasets.

Table 3.2: Dataset Comparison

Comparison of different datasets

Name	Ratings Column	Price Column	Expected update frequency	Usability (out of 10)	Votes	Category
Craigslist Dataset [10]	Yes	Yes	Quarterly	10	1296	Gold
Pakwheels Dataset [11]	No	Yes	Monthly	10	6	None
Cars24 Dataset [12]	No	Yes	Never	8.82	19	None
Geely Auto Dataset [13]	No	Yes	Not specified	7.06	513	Gold

All datasets and their different attributes are taken from Kaggle [14].

3.5 Conclusion

As seen in the previous sections, the car buying and selling processes have certain inefficiencies. There is a minor improvement in the bidding processes around the world. Observations suggest there is a lot of room for improvement in this segment. A review of existing software helped us in developing a deeper understanding. With the help of literature reviews and interacting with existing platforms, we know what features are missing in the previous and current softwares. As a result, Bidding Bazaar is poised to enhance the user experience for both buyers and sellers in the automotive market. We are committed to creating a user-friendly application accessible to anyone.

Chapter 4 Software Requirement Specifications

This chapter contains all the important features of our product. It contains functional and non-functional requirements. It also contains GUI samples, use cases, and database design.

4.1 List of Features

Important features of our system are as under:

- It can register new users
- Our system can perform CURD operations on listings
- It can send notifications when a specific event happens
- It will allow the admin to register more admins
- It will have a payment gateway for buying tokens
- It will allow buyers to bid

4.2 Functional Requirements

Here we have listed down the functional requirements of our project - Bidding Bazaar:

4.2.1 Functional requirements for buyer

Buyers can do the following things:

- Buyer can register on the website
- Buyer can log in to the website
- Buyer can update personal profile
- Buyer can bid on listings
- Buyer can buy tokens to bid on the listing
- Buyer can see listings

4.2.2 Functional requirements for seller

Sellers can do the following things:

- Seller can register on the website

- Seller can log in to the website
- Seller can update personal profile
- Seller can make a listings
- Seller can delete their listings
- Seller can see bids on its listings

4.2.3 Functional requirements for admin

Admin can do the following things:

- Admin can log in to the website
- Admin can make other admins
- Admin can approve or disapprove a listing

4.3 Quality Attributes

The following are some quality attributes:

- System will have high usability. So that people can easily use our system.
- System will have high compatibility. It will enable developers to add modern and latest modules to it.
- System will be scalable. So that it can handle a large number of users.
- System should be maintainable. It will give website more up-time. It will also reduce the efforts of the developers.

4.4 Non-Functional Requirements

The non functional requirements for our final year project are listed below:

4.4.1 Reusability

Our code should be divided into functions for easy maintenance and reusability.

4.4.2 Performance

Our system should be able to complete a request within 5-10 seconds. It should be able to handle a handful of people at same time.

4.4.3 Reliability

Our website should be reliable so that people can rely on it. It should have an uptime of 99 percent.

4.4.4 Security

Ensuring the security of our website is paramount, guaranteeing users a confident and worry-free experience.

4.5 Assumptions

During the design phase, several assumptions were considered, and some of the key ones include:

- Users are expected to possess a reliable and consistent internet connection.
- Users are presumed to have access to a device capable of running web applications.
- Users are anticipated to be familiar with online bidding and payment methods.
- Users are assumed to possess a basic understanding of website navigation and usage.

4.6 Use Cases

This section lists some of the use cases for the project.

4.6.1 User Login

Name	User Login		
Actors	Users (Buyer/Seller)		
Summary	The user will enter their credentials and upon authentication will be taken to the home page.		
Pre-Conditions	The user must not be logged in. And should have signed up.		
Post-Conditions	The user is logged in and taken to the home page.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user opens the login page.	2	The login form is shown and user can enter credentials.
3	The user enters valid email and password.	4	After verification of credentials the user is logged in and taken to the home page.
Alternative Flow			
5	The user enters invalid email or password.	4-A	The system responds with an error notification.

4.6.2 User Sign up

Name	User Sign up		
Actors	User		
Summary	The user will enter their credentials and upon authentication will be taken to the home page.		
Pre-Conditions	The user should not have signed up or be logged in.		
Post-Conditions	User signs up and is taken to the home page.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user opens the sign-up page.	2	Signup form is shown where user can enter credentials.
3	The user enters valid email and password.	4	The system signs up and logs in the user and the user is taken to the home page..
Alternative Flow			
5	The user enters invalid email or password.	4-A	The system responds with an error notification.

4.6.3 User Logout

Name	User Logout		
Actor	Logged-in User		
Summary	Allows a logged-in user to log out.		
Pre-Conditions	The user is logged into the system.		
Post-Conditions	The user is logged out and taken to the login page.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user clicks the ”Logout” button.	2	The system logs the user out and takes them to the login page.

4.6.4 Create Car Listing

Name	Create Car Listing		
Actor	User		
Summary	Allows a registered user to create a listing for a car.		
Pre-Conditions	The user is logged in.		
Post-Conditions	The car listing is pending approval by the admin.		
Special Conditions	None		
Basic Flow			
Actor Action		**System Response**	
1	The user clicks to the ”Add Car” button.	2	The system displays the car listing creation page asking for car details.
3	The user provides the required details for the car listing.	4	The system saves the information.
5	The user clicks on Add button.	6	The system notifies the user that the car listing is pending admin approval.
Alternative Flow			
7	The user provides incorrect details.	4-A	The system after disapproval of the listing request from admin notifies the user.

4.6.5 Listing Deletion

Name	Listing Deletion		
Actor	User		
Summary	Allows a user to delete their car listing from the platform.		
Pre-Conditions	The user is logged in and has a car listing on the platform.		
Post-Conditions	The car listing is deleted.		
Special Conditions	None		
Basic Flow			
Actor Action		**System Response**	
1	The user navigates to their car listing on the platform.	2	The system displays the user's car listing along with the button "Delete".
3	The user click the button "Delete" on their car listing.	4	The system asks the user for confirmation to delete the listing.
4	The user confirms the deletion of the car listing.	6	The system deletes the car listing from the platform and notifies the user of successful deletion.

4.6.6 Listing Updation

Name	Listing Updation		
Actor	User		
Summary	Allows a user to update their car listing on the platform.		
Pre-Conditions	The user is logged in and has a car listing on the platform.		
Post-Conditions	The car listing is updated.		
Special Conditions	None		
Basic Flow			
Actor Action		**System Response**	
1	The user navigates to their car listing on the platform.	2	The system displays the user's car listing along with the button to update.
3	The user clicks the button to update their car listing.	4	The system shows a car listing updation form to the user.
5	The user clicks the "Save" button.	6	The system notifies the user that the car listing is pending admin approval.
Alternative Flow			
7	The user provides incorrect details.	6-A	The system after disapproval of the updation request from admin notifies the user.

4.6.7 Listing Search

Name	Listing Search		
Actor	User		
Summary	Allows a user to search for car listings on the platform.		
Pre-Conditions	The user is logged in.		
Post-Conditions	The results of the search are displayed to the user.		
Special Conditions	None		
Basic Flow			
Actor Action		**System Response**	
1	The user clicks the ”Search Listings” field.	2	The system displays the search interface where the user to search.
3	The user selects search criteria, such as car make, model, or price range.	4	The system retrieves and displays the result of the search
Alternative Flow			
5	The search criteria yield no matching car listings.	4-A	The system notifies the user that no listings match the provided criteria.

4.6.8 Admin Addition

Name	Admin Addition		
Actor	Admin		
Summary	Allows an admin to add a new admin to the platform.		
Pre-Conditions	The admin which is adding has to be logged into the system.		
Post-Conditions	A new admin is added to the platform.		
Special Conditions	None		
Basic Flow			
Actor Action		**System Response**	
1	The admin which is adding navigates to the admin management section.	2	The system displays options to manage admin accounts.
3	The previous admin selects the option to add a new admin.	4	The system provides a form to enter the details of the new admin.
5	The previous admin submits the form with the new admin’s details.	6	The system adds the new admin to the platform and notifies the previous admin of the successful addition.

4.6.9 Admin Deletion

Name	Admin Deletion
Actor	Admin
Summary	Allows an admin to delete another admin from the platform.
Pre-Conditions	The admin performing the deletion is logged in.
Post-Conditions	The specified admin is deleted.
Special Conditions	None

Basic Flow			
Actor Action		**System Response**	
1	The admin performing the deletion navigates to the admin management section.	2	The system displays options to manage admin accounts, including deletion.
3	The admin selects the option to delete an admin and specifies the admin to be removed.	4	The system confirms the admin deletion and notifies the performing admin of the successful removal.

4.6.10 Profile View

Name	User Profile View
Actor	User
Summary	Users can view their information in profile.
Pre-Conditions	The user is logged into the system.
Post-Conditions	The users can view their profile details.
Special Conditions	None

Basic Flow			
Actor Action		**System Response**	
1	The user navigates to ”View Profile” section.	2	The system displays the user’s profile information.

4.6.11 Profile Update

Name	User Profile Update		
Actor	User		
Summary	Allows a user to update their information in profile.		
Pre-Conditions	The user is logged into the system.		
Post-Conditions	Updated user information in profile.		
Special Conditions	None		
Basic Flow			
Actor Action		**System Response**	
1	The user clicks the ”User Profile” button on main screen.	2	The system displays the user information in profile.
3	The user selects the button to update their information in profile.	4	The system takes the user to updation interface.
5	The user provides information and clicks on ”Save” button.	6	The system notify updation successful.
Alternative Flow			
7	The user leaves required fields blank or does not fill in the required format.	6-A	The system prompts users to fill the required fields.

4.6.12 Bid Placement

Name	Bid Placement		
Actor	User		
Summary	Allows a user to place a bid on a car listing.		
Pre-Conditions	The user is logged in and has selected a car listing.		
Post-Conditions	The bid is placed on the car listing.		
Special Conditions	None		
Basic Flow			
Actor Action		**System Response**	
1	The user clicks the car listing they are interested in.	2	The system displays the car listing along with a button to place a bid.
3	The user clicks the button to bid.	4	The system prompts the user to provide their bid and other details.
5	The user submits the bid for the car listing.	6	The system records the bid and notifies the user that the bid has been placed successfully.
Alternative Flow			
7	The user provides an invalid bid amount or details.	6-A	The system prompts the user to correct the input

4.7 Hardware and Software Requirements

The following sub-section of this report contains the software and hardware requirements of our project. Here you can see the list of these requirements.

4.7.1 Hardware Requirements

A computer equipped with at least an Intel Core i5 processor and a minimum of 8 gigabytes of RAM, along with a reliable internet connection, is highly recommended.

4.7.2 Software Requirements

The software which is required for this project is VS code, Windows, Git, Django, React, Node, MongoDB, and one of the latest web browsers.

4.8 Graphical User Interface

GUI sample of the system is as under:



Figure 4.1: Login Page

This illustration displays the login page

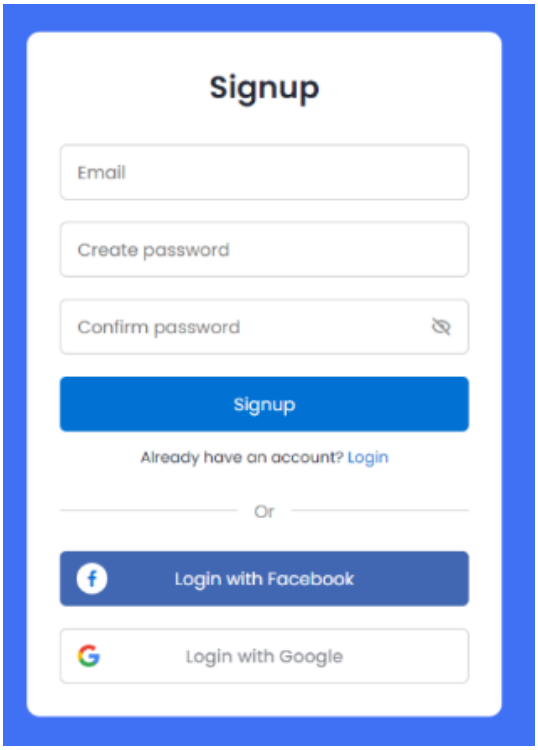


Figure 4.2: User Sign-up Page

This illustration displays the user sign-up page

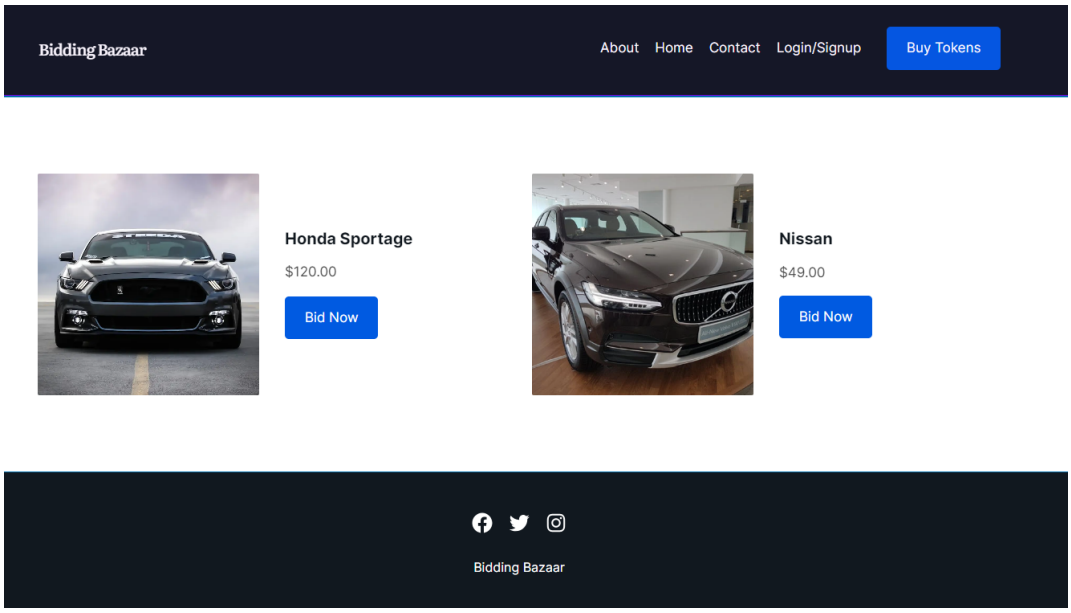


Figure 4.3: Home Page

This illustration displays the home page

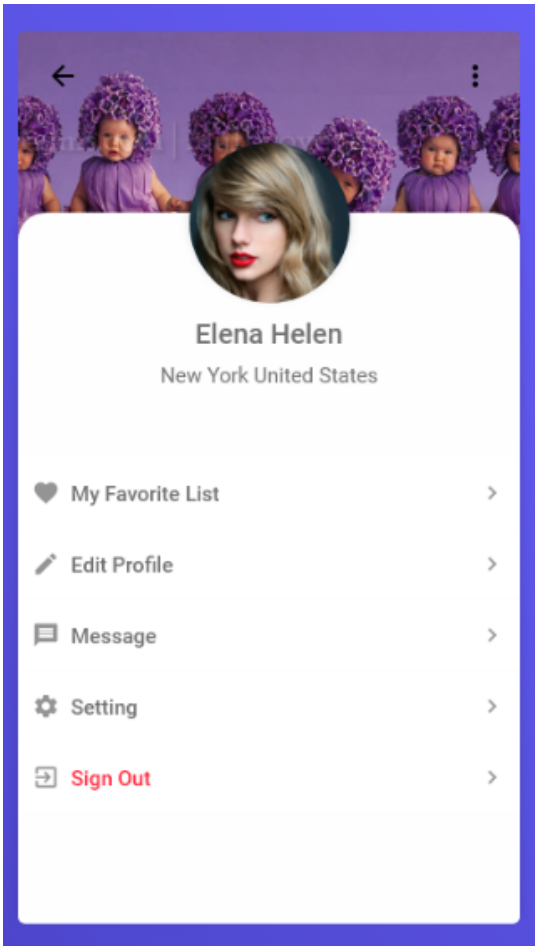


Figure 4.4: User Profile Page

This illustration displays the user profile page

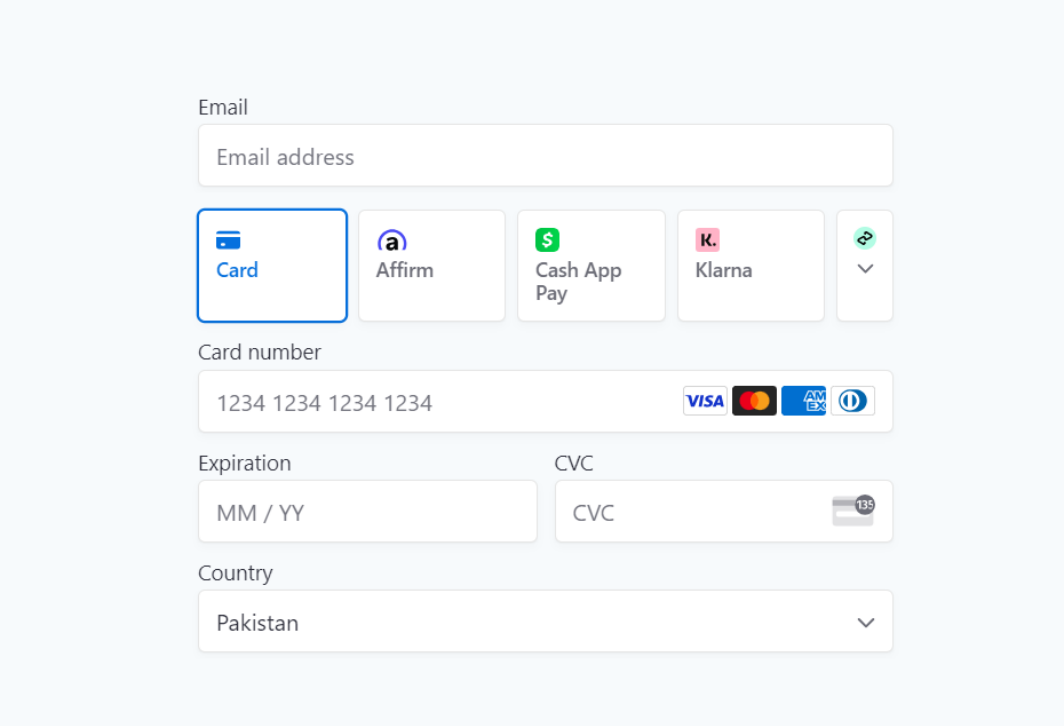


Figure 4.5: Payment Page

This illustration displays the payment page

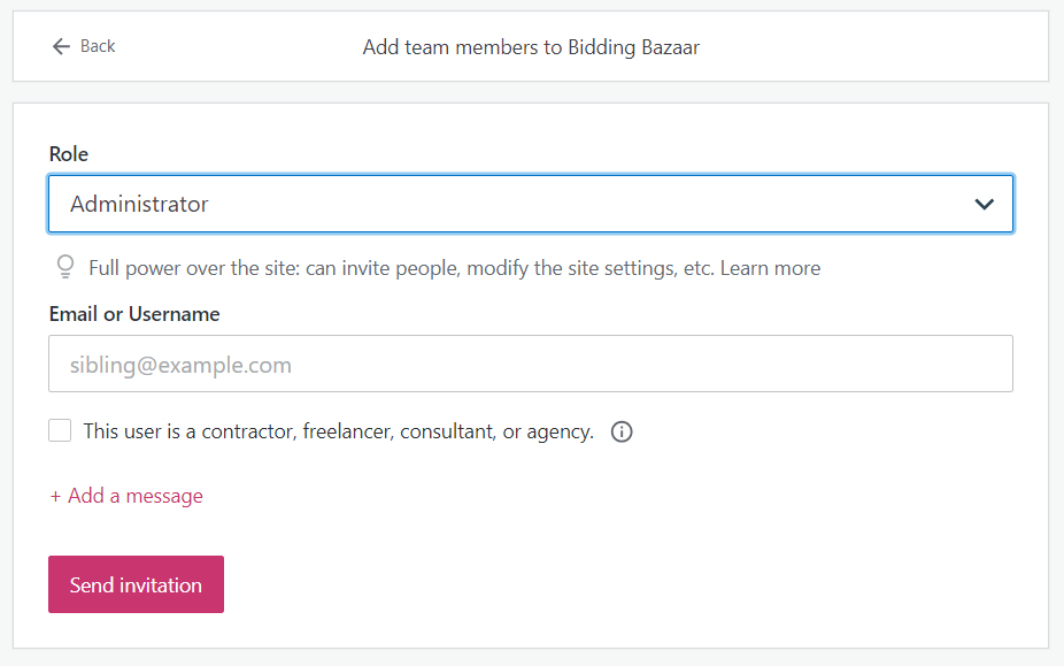


Figure 4.6: Admin Dashboard

This illustration displays the admin dashboard

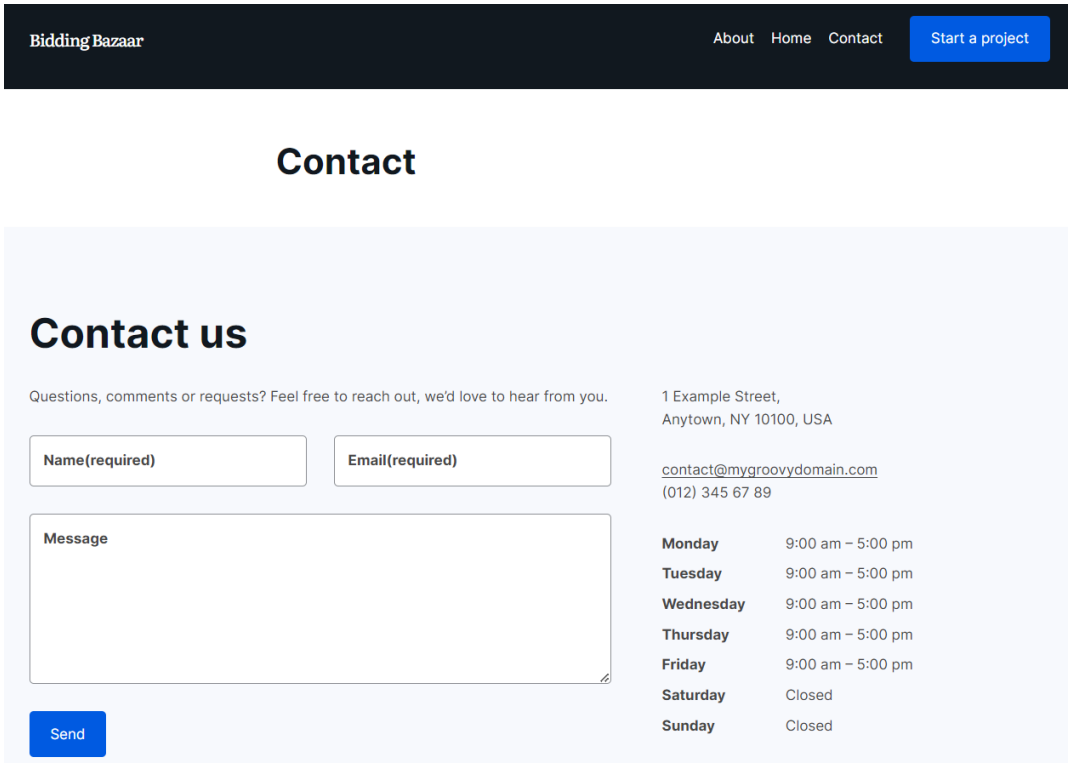


Figure 4.7: Contact-us Page

This illustration displays the contact us page

4.9 Database Design

4.9.1 ER Diagram

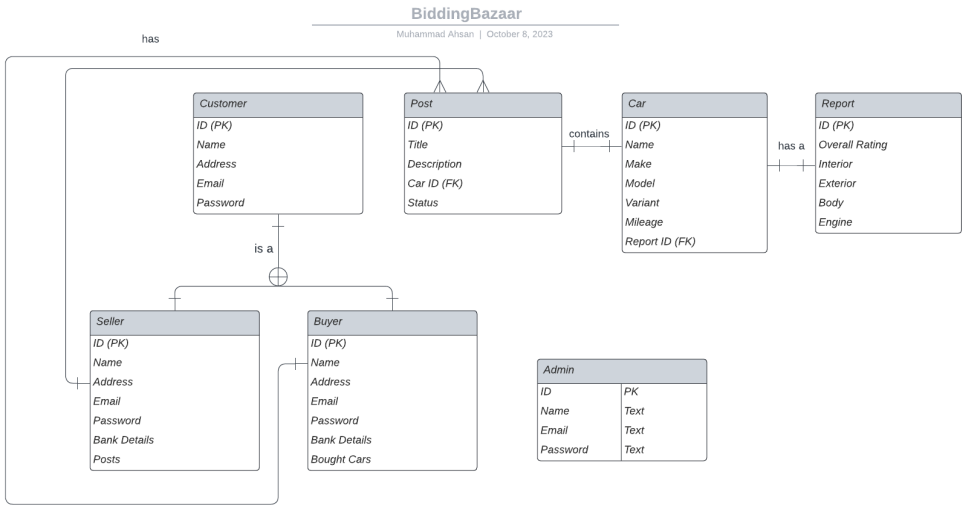


Figure 4.8: ER Diagram

This figure shows the ER Diagram

4.9.2 Data Dictionary

Table 4.1: Table 2: Data dictionary

The table shows the attributes and their data types of different entities

Entity	Attribute	Data Type	Relation To	Nullable	Description
Customer	”ID”	Int		No	Unique ID (PK)
	Name	String		No	User name of customer
	Address	String		No	Address of customer
	Email	String		No	Email of customer
	Password	String		No	Password of customer
	type	String	Seller, Buyer	No	Customer can be a seller, buyer or both
Seller	Bank Details	String		No	Bank Account details of customer
	Posts	Post	Post	No	Posts created by the seller
Buyer	Bank Details	String		No	Bank Account details of customer
	Bought Cars	Post		No	Cars bought by a buyer
Post	”ID”	Int		No	Unique ID (PK)
	Title	String		No	Title of the auction post
	Description	String		No	Description of the car
	Status	String		No	Sold or not
	Car ID	Int	Car	No	Unique ID (FK)
Car	”ID”	Int		No	Unique ID (PK)
	Name	String		No	Name of the car
	Make	String		No	Year of make
	Model	String		No	Model of the car
	Variant	String		No	Variant of the car
	Mileage	Int		No	Mileage on the car
	Report ID	Int	Report	No	ID of the report(FK)
Report	”ID”	Int		No	Unique ID (PK)
	Rating	Float		No	Overall rating
	Interior	Float		No	Interior rating
	Exterior	Float		No	Exterior rating
	Body	Float		No	Body rating
	Engine	Float		No	Engine condition rating
Admin	”ID”	Int		No	Unique ID (PK)
	Name	String		No	Name of the admin
	Email	String		No	Email of admin
	Password	String		No	Password of admin

4.10 Risk Analysis

Following is the list of risks that can be encountered during the project:

4.10.1 Data for Price Prediction

We will use data from external sources. This data should be accurate to predict accurate results. Inaccurate data will affect our results.

4.10.2 Performance

The platform performance may be affected if a large number of users try to access it simultaneously.

Chapter 5 High-Level and Low-Level Design

5.1 System Overview

In our system, the user will sign up and then log in to our system. The user can then request to post his car for listing. A form will appear prompting the user to add the car details. After the details are submitted it will be sent to the Python server, which will generate a price based on the features and condition of the car. After that, the car images will be sent to another Python server to recognize if the listing being posted is actually a car or not. On a true response from the server, the car's listing will be posted on the platform and saved on the database server. The buyer who is also a user can fetch the listing from the database server.

5.2 Design Considerations

Here in this section, we will address the complete design solution.

5.2.1 Assumptions and Dependencies

While designing this system, some of the assumptions that we are making are listed below:

- We are assuming that the user has a computer or smart phone for accessing the website.
- User has knowledge of Basic English for using this website.
- For accessing this website, user has an internet connection.
- User has a basic knowledge on how to access general websites.
- Modern and good browsers

5.2.2 General Constraints

5.2.2.1 Hardware or software environment

- The project assumes the user to have an OS better than Windows 7 or equal.
- A RAM greater than 4GB or equal.
- Stable internet connection.

5.2.2.2 End-user environment

- The user runs the website on a known browser. Some are Chrome, Firefox, Safari and Microsoft edge.
- Knowledge of how to use a browser and internet.

5.2.2.3 Availability or volatility of resources

- We assume that users would have the resources to run the website.
- Must have a computer, a browser, and a stable internet connection.

5.2.2.4 Standards compliance

- VS code for development
- MongoDB backend
- JS libraries

5.2.2.5 Interface/protocol requirements

- Frontend will be in React.
- The interface will run on Javascript supported browsers.
- Https will be used.

5.2.2.6 Data repository and distribution requirements

- Mongo DB will be used for data repository and distribution requirements.

5.2.2.7 Security requirements (or other such regulations)

- The system will ensure to make the passwords of users strong.
- All the data will be backed up regularly.

5.2.2.8 Language Constraints

Only the English language will be used for the web application.

5.2.2.9 Security Requirements

Any of the three key requirements such as confidentiality, integrity, and availability, which support one another cannot be fully disregarded, regardless of security policy objectives. Confidentiality, for instance, is required to safeguard passwords. So, we will provide all the above-mentioned properties.

5.2.3 Goals and Guidelines

The goals and guidelines on which our system is built are given below:

5.2.3.1 User Friendly

Our system will be user friendly as it will have a better UI and UX than other websites of the same domain. In addition to this our website will be in English which will be very helpful.

5.2.3.2 Performance

Our system will be a high performance system. It will not require unnecessary memory or space from the end user. It will perform all operations with very little latency.

5.2.3.3 Modifiability

Our system will be modifiability so that new features can be easily added according to the future needs.

5.2.4 Development Methods

Scrum will be used for development. Our work will be divided into sprints, which will have deadlines. These sprints will be further divided into individual tasks, for this, we will use software like Kanban board or Trello. The period for each sprint will be at max one week. After the completion, we will have a short meeting. In these meetings, we will discuss the work, review the progress, and schedule the rest of the work.

5.3 System Architecture

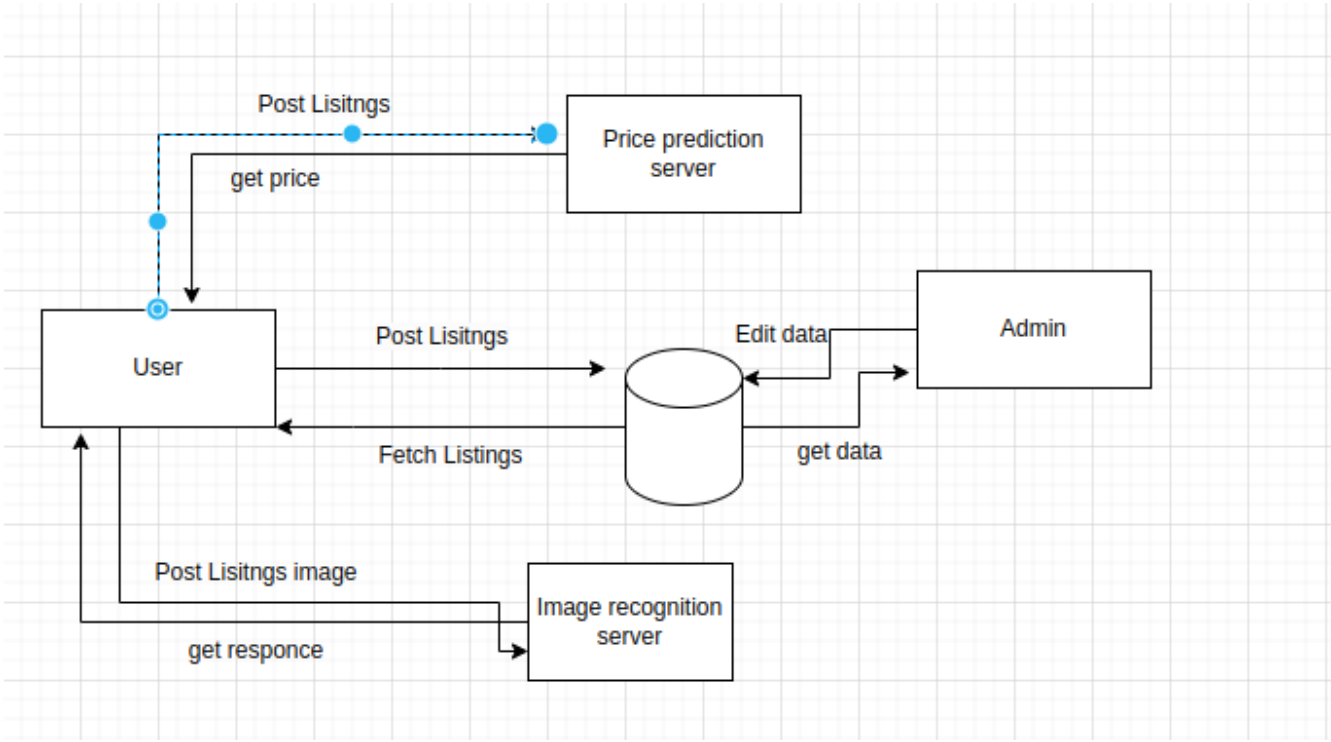


Figure 5.1: Architecture Diagram

System's high level diagram

As in the diagram, our system will have a central database. There are two Python servers. One server is to predict the price when the user posts a listing, the other is to identify if the listing is a car. The user sends request to both these servers before posting the car to the central database. The Admin module can access the central database and update and retrieve information.

5.3.1 Subsystem Architecture

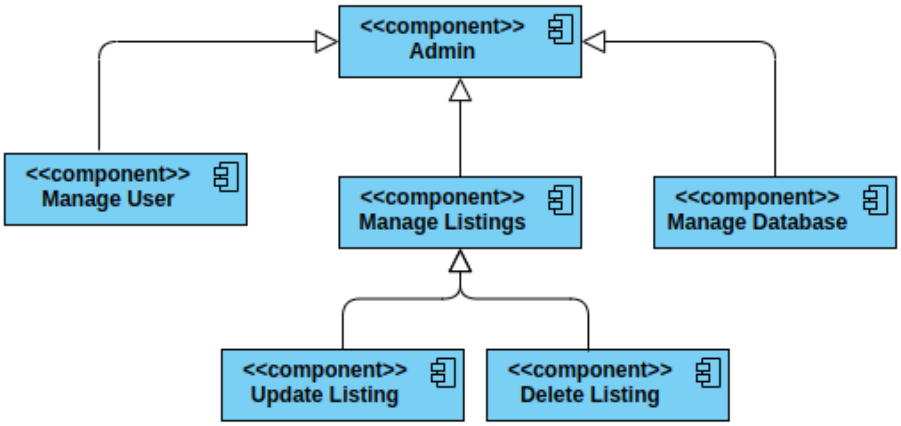


Figure 5.2: Admin Component Diagram

The internal view of admin component is given above

As shown in the Admin component diagram. The Admin can Manage Users, Manage the listing, and Manage Database. The Admin will have components to access these settings.

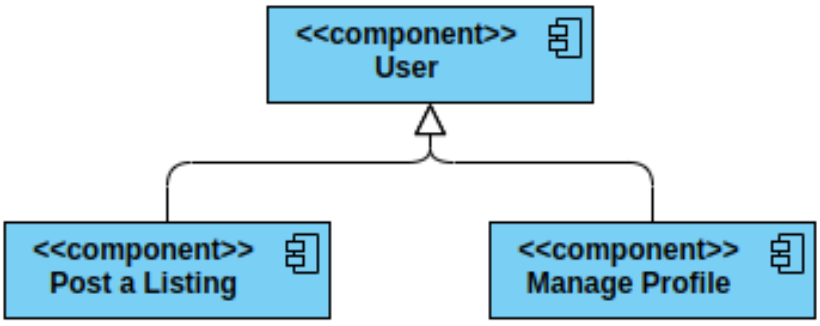


Figure 5.3: User Component Diagram

The internal view of user component is given above

As shown in the User component diagram. The can post listings, retrieve them, and manange his profile. A module for each will be created.

5.4 Architectural Strategies

Following are the architectural strategies we will be using for this project:

5.4.1 Programming Language

The system shall be using MERN stack technology. The front end will be built in React js and the back end in Node js. We will be using Python for price prediction and the car verification system. For the database, we will be using Mongo DB. We are using the MERN stack as it is better in terms of web development than others as it provides component-based design. It has a robust framework and a huge library collection, saving a lot of time. React js allows for rapid development. Also, the choice of Python is based on the reason that Python has a huge collection of libraries having pre-trained models for prediction and computer vision.

5.4.2 User-Interface Paradigm

We will be using Schneider man's Eight Golden Rules for the user interface. They will guide the styling and help to enhance the user experience. Using these rules ensures user interface is user-friendly and effortless. for users

5.4.3 Concurrency and Synchronization

Async operations will be used for back-end queries. This will enable the user to interact with the front end while the back-end system is working.

5.4.4 Error Detection and Recovery

Detecting and handling errors in both front-end and back-end will ensure that the system is reliable. Meaningful error messages will be displayed to the user. Any errors will be added to the logs for analysis.

5.4.5 Hosting and Version Control

GitHub will be used for hosting our repository and for version control we will be using git. Version control is important as it will keep the team synchronized and updated.

5.5 Domain Model/Class Diagram

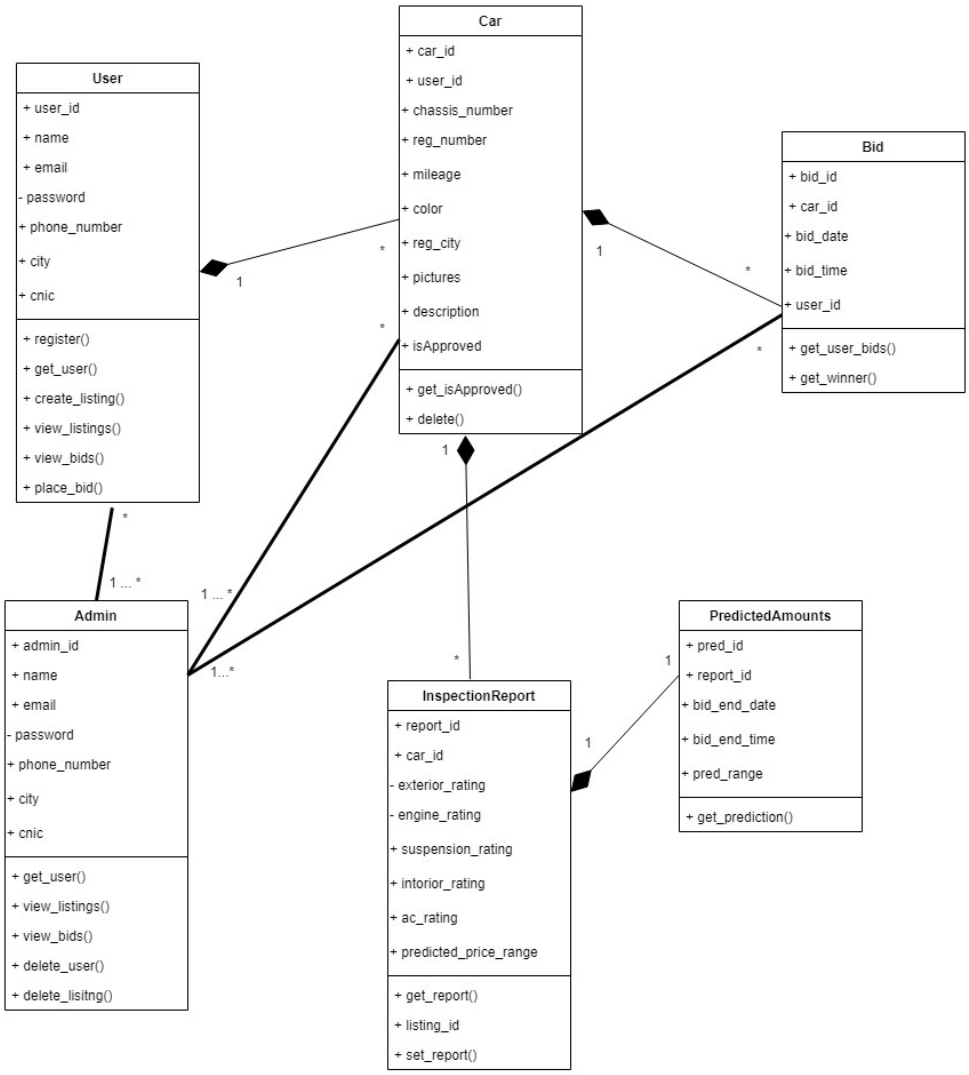


Figure 5.4: Class Diagram

The class diagram of system is given above

5.6 Sequence Diagrams

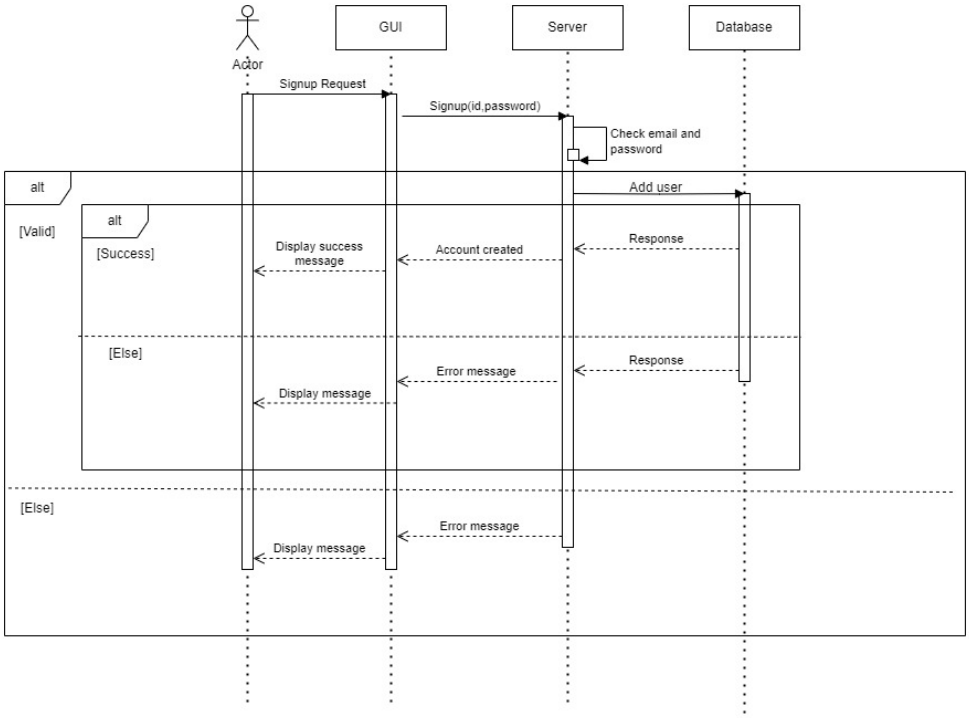


Figure 5.5: Sign up

The sequence diagram of sign up is given above

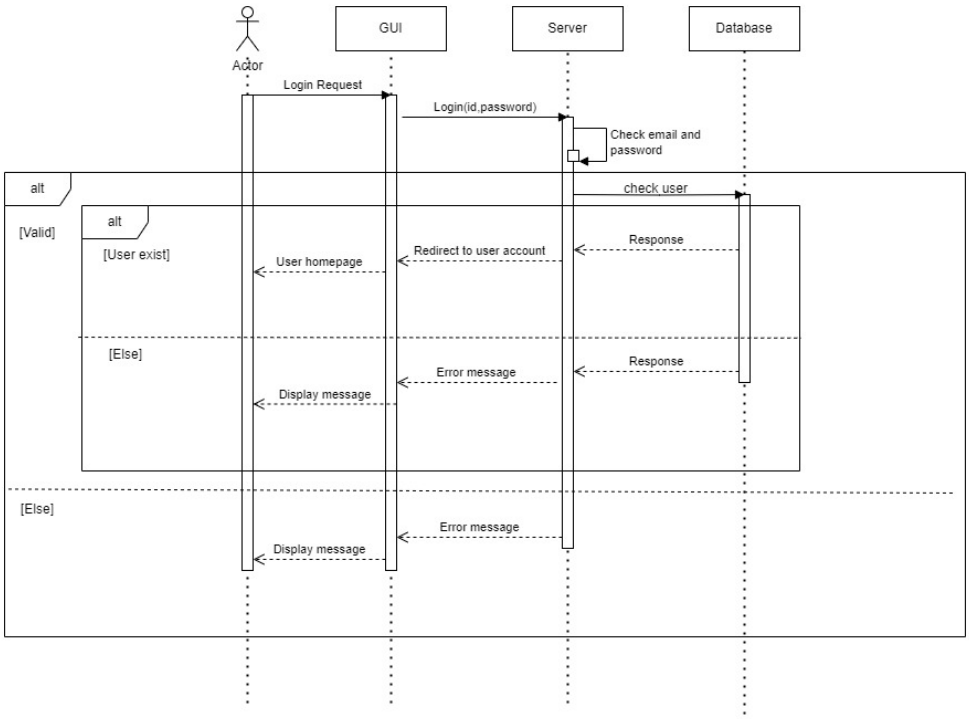


Figure 5.6: Login

The sequence diagram of login is given above

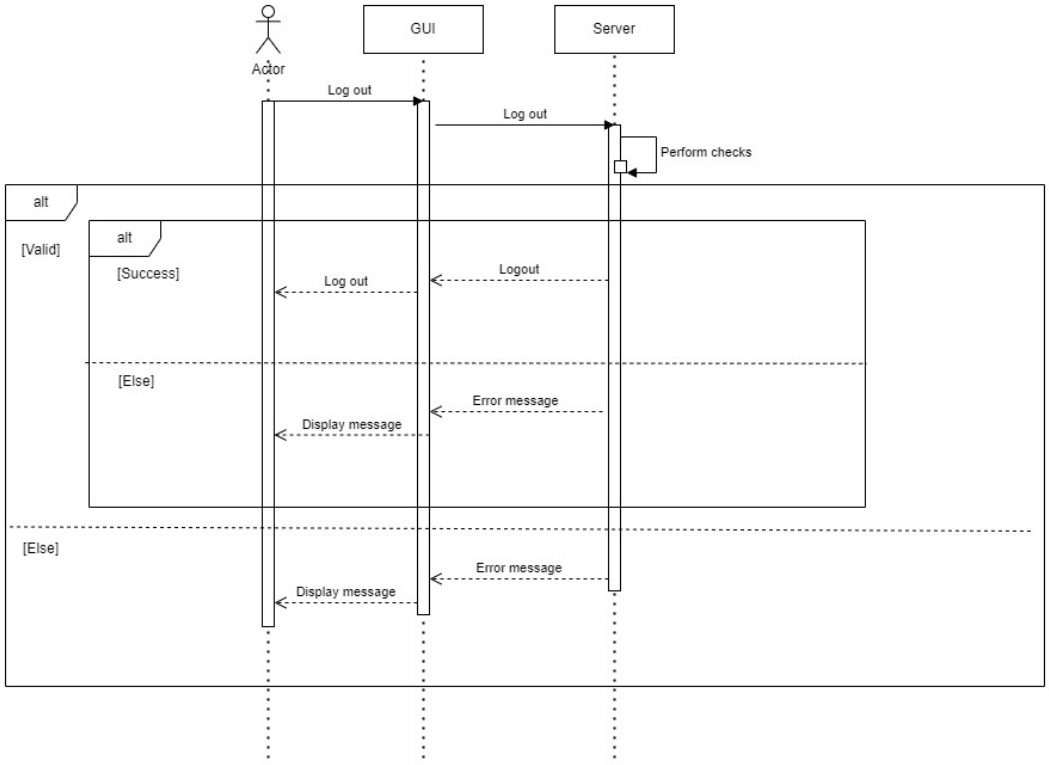


Figure 5.7: Log out

The sequence diagram of log out is given above

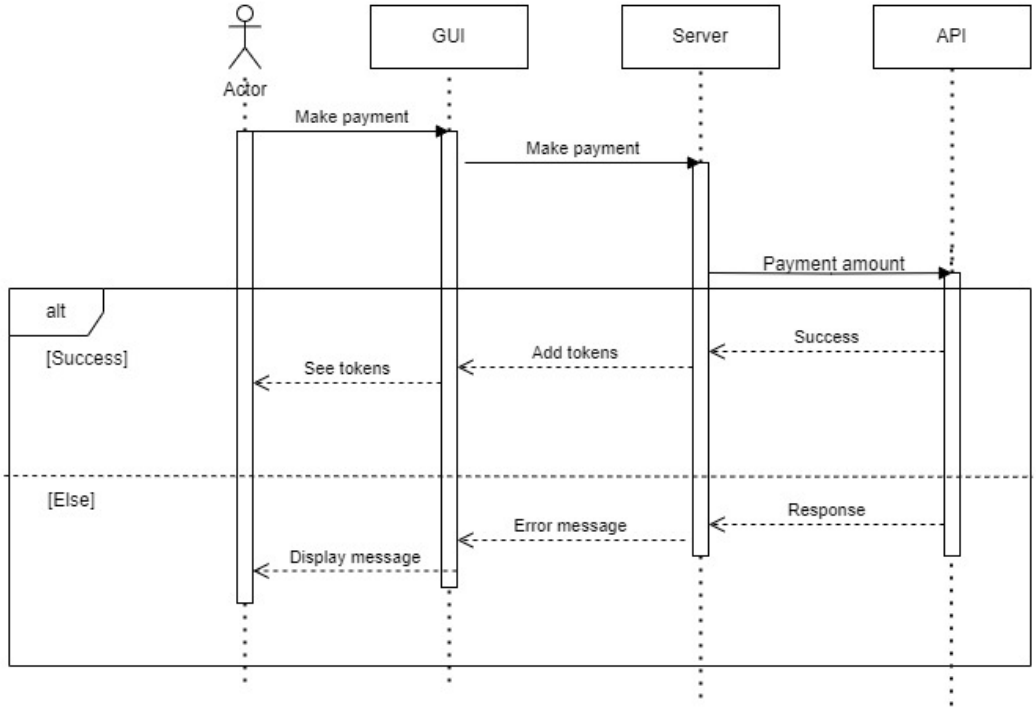


Figure 5.8: Payment

The sequence diagram of payment is given above

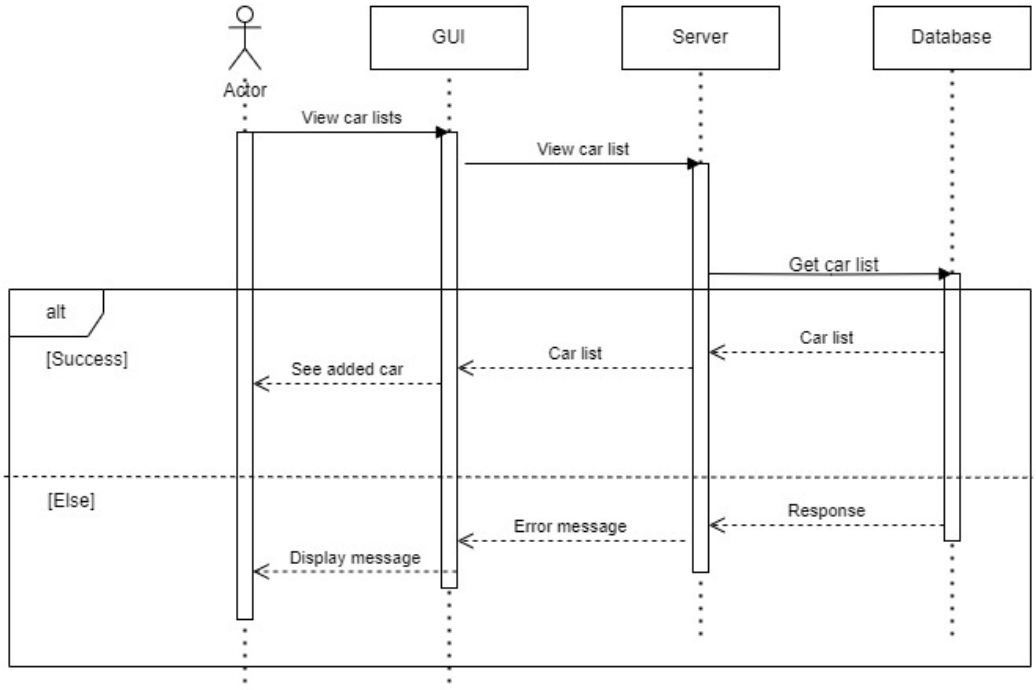


Figure 5.9: View cars

The sequence diagram of view cars is given above

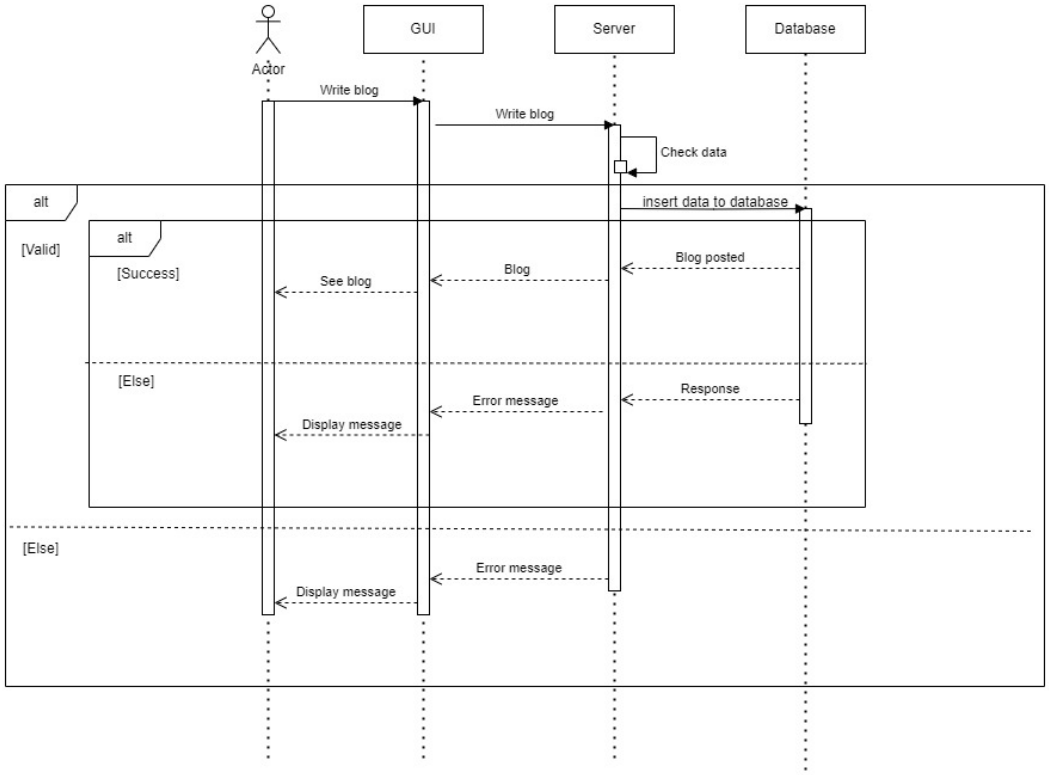


Figure 5.10: Post blog

The sequence diagram of post blog is given above

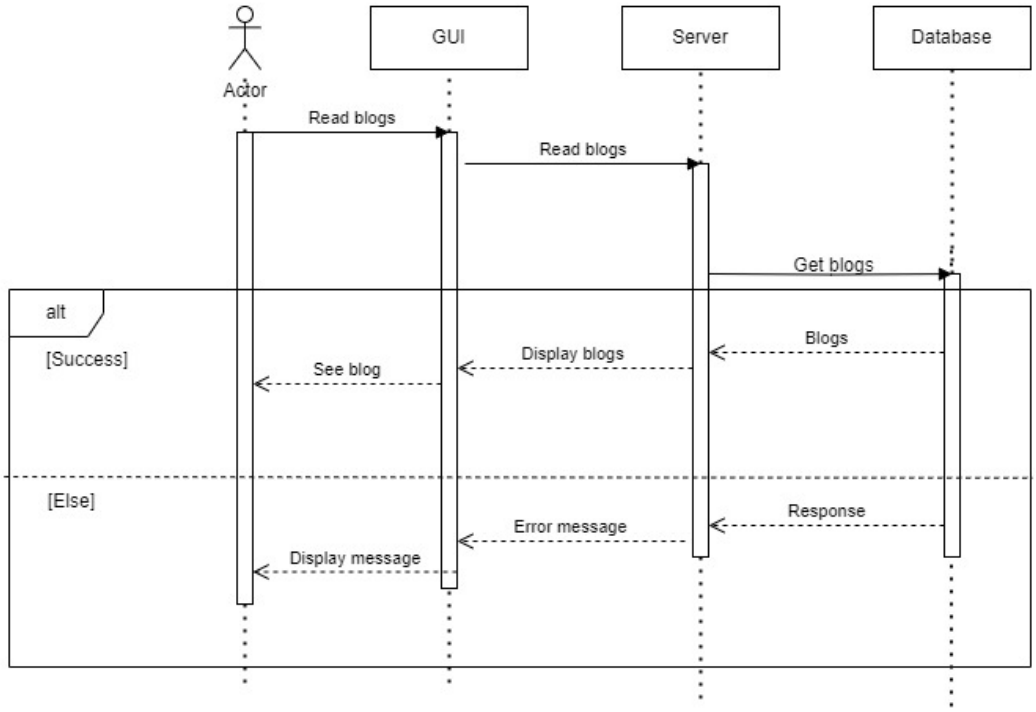


Figure 5.11: Read blog

The sequence diagram of read blog is given above

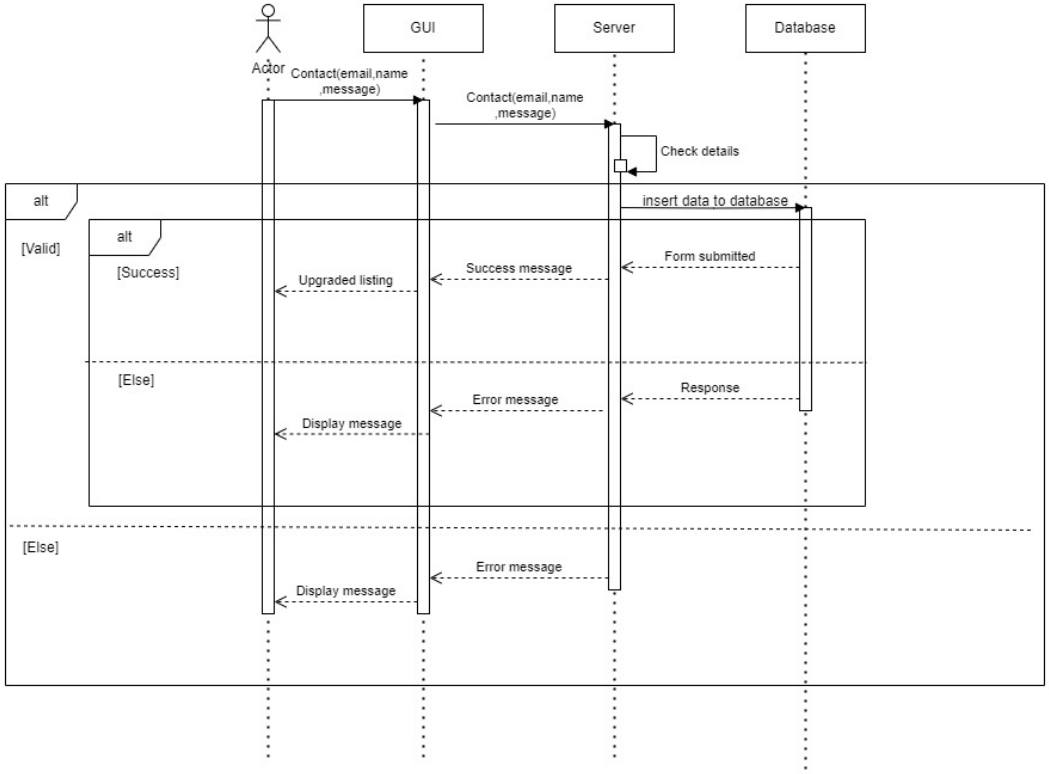


Figure 5.12: Contact us

The sequence diagram of contact us is given above

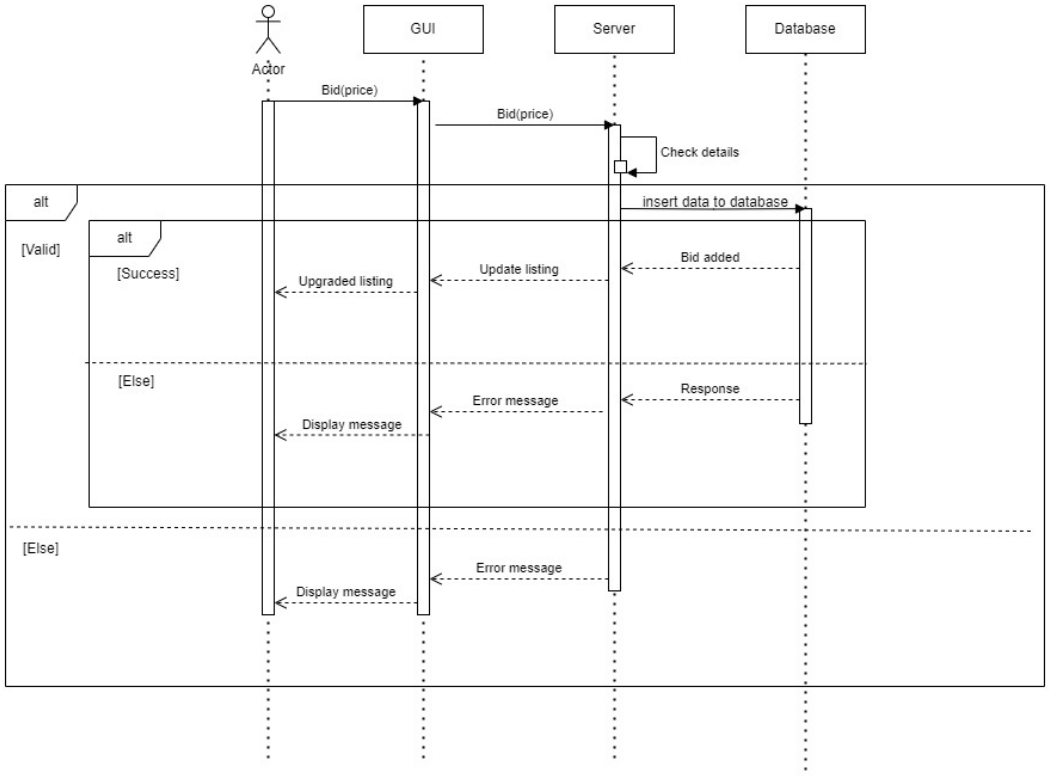


Figure 5.13: Bidding

The sequence diagram of bidding is given above

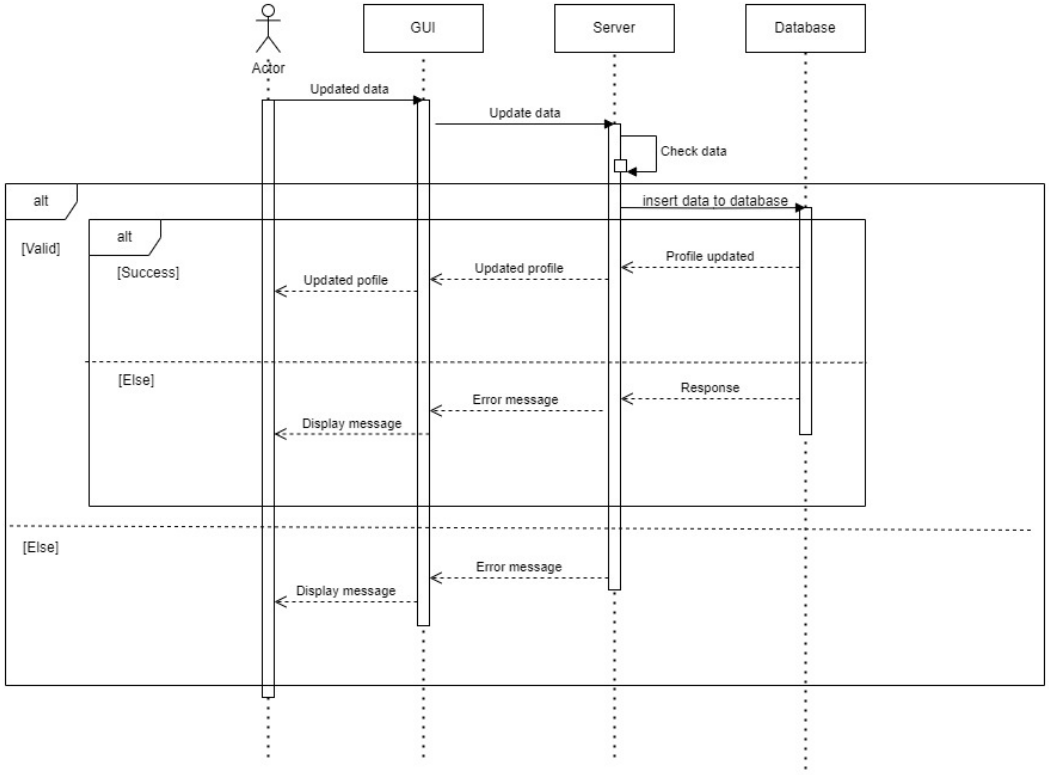


Figure 5.14: Update profile

The sequence diagram of update profile is given above

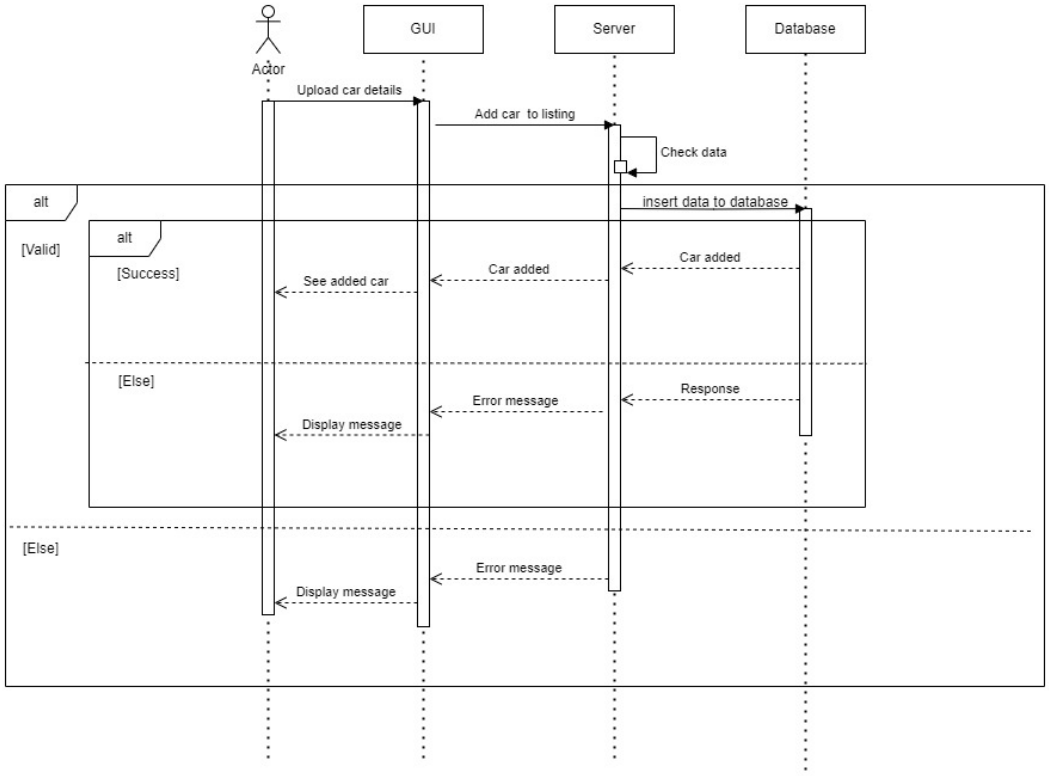


Figure 5.15: Post car

The sequence diagram of post car is given above

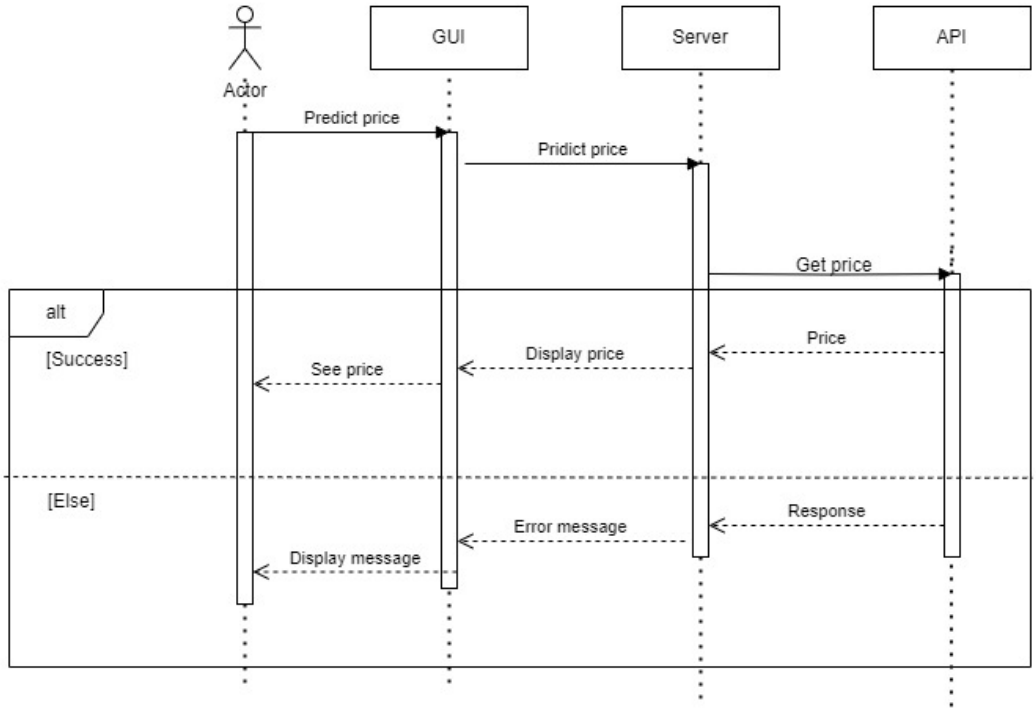


Figure 5.16: Predict price

The sequence diagram of predict price is given above

5.7 Policies and Tactics

5.7.1 Conventions and Coding Guidelines

Proper coding guidelines regarding MERN stack and Python will be used so the code is easily understandable. Proper programming conventions will be used. Proper comments in the code will be added for easy understanding and future maintenance. This will give a proper flow for the project.

5.7.2 Coding Environment

The coding environment we will be using for our project is VS code. As it has an editor, compiler, and interpreter all together built into it. It also has huge support, tools, packages, and extensions. It has an auto-indentation feature and also provides suggestions for syntax which makes development easy.

5.7.3 System Testing

We will be using both white and black box techniques for testing different modules of the project. Before deploying, we will thoroughly test every aspect to avoid any errors.

5.7.4 Maintenance of Software

After the completion of the project, we will be regularly updating the project and adding new features as required.

5.7.5 Extensibility

The system's extensibility will be kept in mind. This will allow us to add new features in the future. For this, we will use a modular approach and microservices for every part.

Chapter 6 Implementation and Test Cases

All the details related to the components built or added in the Bidding Bazaar prototype are described below.

6.1 Implementation

The prototype was developed in the MERN stack. We have implemented important functionalities for the prototype. To better present our idea in the presentation, we have implemented all the important frontend pages. The frontend pages included in the prototype are the main page, login/signup pages, user profile page, contact us page, car page, add car page, admin login, and team page. To make our prototype functional, we have added middleware and backend. Implementation details for the Bidding Bazaar prototype components will be provided in the sub-sections below.

6.1.1 Authentication

One of the most crucial parts of any system is its authentication process. It is a crucial part as it maintains the confidentiality and integrity of the system. We have provided email and google authentication options to the user through the Firebase APIs. By using Firebase API, user credentials will be stored in Firebase and the rest of the user information will be stored in MongoDB. Upon login or signup, these APIs will return an ID token. This ID token will be used as an access token by our backend as it can be verified by sending it to Firebase and decoding it will provide us with a user ID. On request, we can then show protected information according to this user ID. For sign-out again we have used Firebase's signout API.

6.1.2 Bidding

The main and most prominent feature of Bidding Bazaar is the bidding process. In our prototype, we have implemented this feature. Whenever a buyer places a bid, information is posted to the database, and for all other users, their frontend will send requests every 3 seconds to fetch any new bids. This way all the bidding data we be kept up to date among all the users.

6.1.3 Car Detection

To streamline the process of car auction and to make it efficient, we will be verifying all the car images uploaded by the user to check whether they are genuine or fraudulent. We can do this by checking if the images contain the same model of car as the user have provided. For this purpose, we have created a Python script that uses a pre-trained model yolov8, imported from the Yolo library, and custom trained it

on models of cars which are in Pakistan. This automatic car image detection will filter out all irrelevant posts and will make the listing process efficient.

6.1.4 Scraper

A scraper has been developed using the Python library Beautiful Soup to gather data for our prediction model. It stores all scraped data in a comma-separated file (CSV). The scraper begins by requesting the URL of the desired page on a website. When the scraper makes a request, it fetches the publicly available HTML code from the specific page. The scraper extracts data from the values present in the HTML tags. These tags are chosen according to our needs. We only scrape those tags which we need, while ignoring others. To store the scrapped values, it first checks for the existence of a CSV file. If an old CSV file exists, it appends new data to it; otherwise, it creates a new file. The scraper extracts the total number of pages from the site before commencing its scraping process. If the targeted website is updated, the scraper also needs to be changed based on the new website structure.

6.1.5 AI Recommendation

To enhance the user experience, we have implemented user-based content filtering for recommendation. It provides recommendations based on popular car postings among other users who have interacted with the same car as the current user. With this recommendation system, users will be shown relevant content in which they might be interested, thereby enhancing the overall user experience.

6.1.6 Price Prediction

Our price prediction model will utilize the data scraped by our dedicated scraper. Before feeding the data into our model, we'll undertake dataset cleaning. Pandas is mainly used for data cleaning. The price prediction model will be accessible via a Python-based API. API and python server is created on Flask. It will take car details as input and, after making predictions, return the forecasted price as output. For this purpose we compared the results of 8 different pre-trained models with 3 different types of encoding schemes. These models included XGBoost, Decision Tree, Random Forest, Gradient Boosting, K-Nearest Neighbors, Simple Regression, Support Vector, and Artificial Neural Network. We tested all of these models with one-hot encoding, binary encoding, and label encoding. After conducting practical work, we discovered that combining the XGBoost model with label encoding yielded the best results.

6.1.7 Admin Panel and Website Metrics

After logging in, the admin can add new admins, write blog posts, and delete listings. Admin can also view the system related information like CPU usage, memory usage etc through admin panel. Additionally, website metrics can be viewed through Google Analytics.

6.2 Test case Design and description

Following test cases were developed, designed and conducted in order to check Bidding Bazaar platform for abnormalities, errors and irregularities.

Table 6.1: User Login Test Case

Test case details for user login functionality

User Login			
Test Case ID:	*1*	**QA Test Engineer:**	*Abdullah Sohail*
Test case Version:	*1*	**Reviewed By:**	*Muhammad Ahsan*
Test Date:	*01-04-2024*	**Use Case Reference(s):**	*User Login use case (1)*
Revision History:	*None*		
Objective:	*To test whether the user can log in to the website successfully*		
Product/Ver/ Module:	*Login Module of Website*		
Environment:	*Internet Connection and Website running*		
Assumptions:	*User has already signed up on the website*		
Pre-Requisite:	*The user has already registered and is stored in database.*		
Step No.	**Execution description**	**Procedure result**	
1	*Enter email and password. Click on login button.*	*User is redirected to homepage.*	
Comments: The test case is passed. Our system is working according to our need.			
■ *Passed* ☐ *Failed* ☐ *NotExecuted*			

Table 6.2: User Signup Test Case

Test case details for user signup functionality

User Signup			
Test Case ID:	*2*	**QA Test Engineer:**	*Abdullah Sohail*
Test case Version:	*1*	**Reviewed By:**	*Muhammad Ahsan*
Test Date:	*01-04-2024*	**Use Case Reference(s):**	*User Signup use case (2)*
Revision History:	*None*		
Objective:	*To test whether the user can signup to the website successfully*		
Product/Ver/ Module:	*Signup Module of Website*		
Environment:	*Internet Connection and Website running*		
Assumptions:	*User have a valid email and phone number*		
Pre-Requisite:	*The user is on signup page and have a valid email and phone number.*		
Step No.	**Execution description**	**Procedure result**	
1	*Fill the signup form. Click on signup button. Verify your email.*	*System asks for email verification. Signup successful email received.*	
Comments: The test case is passed. Our system is working according to our need.			
■ *Passed* ☐ *Failed* ☐ *NotExecuted*			

Table 6.3: User Logout Test Case

Test case details for user logout functionality

User Logout			
Test Case ID:	*3*	**QA Test Engineer:**	*Abdullah Sohail*
Test case Version:	*1*	**Reviewed By:**	*Muhammad Ahsan*
Test Date:	*01-04-2024*	**Use Case Reference(s):**	*User Logout use case (3)*
Revision History:	*None*		
Objective:	*To test whether the user can logout from the website successfully*		
Product/Ver/ Module:	*Logout Module of Website*		
Environment:	*Internet Connection and Website running*		
Assumptions:	*User is already logged in*		
Pre-Requisite:	*The user has already login*		
Step No.	**Execution description**	**Procedure result**	
1	*Click on logout button in the navbar.*	*A pop showed. Logout button disappears. Signup and login buttons are available now.*	
Comments: The test case is passed. Our system is working according to our need.			
■ *Passed* ☐ *Failed* ☐ *NotExecuted*			

Table 6.4: New Listing Test Case

Test case details for new listing functionality

New Listing			
Test Case ID:	*4*	**QA Test Engineer:**	*Abdullah Sohail*
Test case Version:	*1*	**Reviewed By:**	*Muhammad Ahsan*
Test Date:	*01-04-2024*	**Use Case Reference(s):**	*New Listing use case (4)*
Revision History:	*None*		
Objective:	*To test whether the user can create a new listing on the website successfully*		
Product/Ver/ Module:	*Create Listing Module of Website*		
Environment:	*Internet Connection and Website running*		
Assumptions:	*User is logged in*		
Pre-Requisite:	*The user have uploaded only car pictures*		
Step No.	**Execution description**	**Procedure result**	
1	*Fill the listing form. Click on post button.*	*A pop will be shown above. Listing is now available for bidding.*	
Comments: The test case is passed. Our system is working according to our need.			
■ *Passed* ☐ *Failed* ☐ *NotExecuted*			

Table 6.5: Post Delete Test Case

This test case details for Car Post Deletion.

Post Delete			
Test Case ID:	*5*	**QA Test Engineer:**	*Usman Ghani*
Test case Version:	*1*	**Reviewed By:**	*Abdullah Sohail*
Test Date:	*01-04-2024*	**Use Case Reference(s):**	*Listing Deletion Use Case (5)*
Revision History:	*None*		
Objective:	*To test whether the user can delete their car posts successfully.*		
Product/Ver/ Module:	*Listing Module of Website*		
Environment:	*Internet is connected, and the website is running.*		
Assumptions:	*User have at least one post.*		
Pre-Requisite:	*User must be logged in to the platform.*		
Step No.	**Execution description**	**Procedure result**	
1	*Click "Delete" button, on the car post.*	*System sends notification that post is deleted successfully.*	
Comments: The test case is passed and user is successfully able to delete their posts.			
■ *Passed* ☐ *Failed* ☐ *NotExecuted*			

Table 6.6: Post Update Test Case

This test case details for Post Updation.

Post Update			
Test Case ID:	*6*	**QA Test Engineer:**	*Usman Ghani*
Test case Version:	*1*	**Reviewed By:**	*Abdullah Sohail*
Test Date:	*01-04-2024*	**Use Case Reference(s):**	*Listing Updation Use Case (6)*
Revision History:	*None*		
Objective:	*To test whether the user can update their car posts successfully.*		
Product/Ver/ Module:	*Listing Module of Website*		
Environment:	*Internet is connected, and the website is running.*		
Assumptions:	*User have at least one post.*		
Pre-Requisite:	*User must be logged in to the platform.*		
Step No.	**Execution description**	**Procedure result**	
1	*Click "Update" button, on the car post*	*System sends notification that post is updated successfully*	
Comments: The test case is passed and user is successfully able to update their posts.			
■ *Passed* ☐ *Failed* ☐ *NotExecuted*			

Table 6.7: Post Search Test Case

This test case details for Post Searching.

Post Search			
Test Case ID:	*7*	**QA Test Engineer:**	*Usman Ghani*
Test case Version:	*1*	**Reviewed By:**	*Abdullah Sohail*
Test Date:	*01-04-2024*	**Use Case Reference(s):**	*Listing Search Use Case (7)*
Revision History:	*None*		
Objective:	*To test whether the user can search car posts properly.*		
Product/Ver/ Module:	*Listing Module of Website*		
Environment:	*Internet is connected, and the website is running*		
Assumptions:	*Platform has atleast one car posted.*		
Pre-Requisite:	*None*		
Step No.	**Execution description**	**Procedure result**	
1	*Click "Search", on top of main page.*	*System shows filtered result based on the key words.*	
Comments: The test case is passed and user is properly able to search through posts.			
■ *Passed* ☐ *Failed* ☐ *NotExecuted*			

Table 6.8: Bid Placement Test Case

This test case details for the bidding feature.

Bid Placement			
Test Case ID:	*8*	**QA Test Engineer:**	*Muhammad Ahsan*
Test case Version:	*1*	**Reviewed By:**	*Usman Ghani*
Test Date:	*01-04-2024*	**Use Case Reference(s):**	*Bid Placement (12)*
Revision History:	*None*		
Objective:	*To test whether the user can place bid on a listing successfully*		
Product/Ver/ Module:	*Bidding Module of Website*		
Environment:	*Internet Connection and Website running*		
Assumptions:	*None*		
Pre-Requisite:	*The user has already registered and can participate in bidding*		
Step No.	**Execution description**	**Procedure result**	
1	*The user clicks Bid button on listing*	*The user's bid gets added to the listing*	
Comments: The test case is passed. Our system is working according to our need.			
■ *Passed* ☐ *Failed* ☐ *NotExecuted*			

Table 6.9: Admin Deletion Test Case

This test case details for deletion of an admin from the system.

Profile View			
Test Case ID:	*9*	**QA Test Engineer:**	*Muhammad Ahsan*
Test case Version:	*1*	**Reviewed By:**	*Usman Ghani*
Test Date:	*01-04-2024*	**Use Case Reference(s):**	*Admin Deletion (9)*
Revision History:	*None*		
Objective:	*To test whether an admin can delete other admin.*		
Product/Ver/ Module:	*Admin Module of Website*		
Environment:	*Internet Connection and Website running*		
Assumptions:	*None*		
Pre-Requisite:	*The admin has already registered and is stored in database.*		
Step No.	**Execution description**	**Procedure result**	
1	*Admin selects the admin and clicks delete*	*The admin gets deleted*	
Comments: The test case is passed. Our system is working according to our need.			
■ *Passed* ☐ *Failed* ☐ *NotExecuted*			

Table 6.10: Profile View Test Case

This test case details for user view of his profile.

Profile View			
Test Case ID:	*10*	**QA Test Engineer:**	*Muhammad Ahsan*
Test case Version:	*1*	**Reviewed By:**	*Usman Ghani*
Test Date:	*01-04-2024*	**Use Case Reference(s):**	*Profile View (10)*
Revision History:	*None*		
Objective:	*To test whether the user can view his profile successfully*		
Product/Ver/ Module:	*User profile Module of Website*		
Environment:	*Internet Connection and Website running*		
Assumptions:	*None*		
Pre-Requisite:	*The user has already registered and is stored in database.*		
Step No.	**Execution description**	**Procedure result**	
1	*The user clicks Profile button at top right*	*The user is redirected to profile page.*	
Comments: The test case is passed. Our system is working according to our need.			
■ *Passed* ☐ *Failed* ☐ *NotExecuted*			

Table 6.11: Profile Update Test Case

This test case details for updating the user profile.

Profile Update			
Test Case ID:	*11*	**QA Test Engineer:**	*Muhammad Ahsan*
Test case Version:	*1*	**Reviewed By:**	*Usman Ghani*
Test Date:	*01-04-2024*	**Use Case Reference(s):**	*Profile Update (11)*
Revision History:	*None*		
Objective:	*To test whether the user can update his profile successfully*		
Product/Ver/ Module:	*User profile Module of Website*		
Environment:	*Internet Connection and Website running*		
Assumptions:	*None*		
Pre-Requisite:	*The user has already registered and is stored in database.*		
Step No.	**Execution description**	**Procedure result**	
1	*The user clicks update button at profile page*	*The user updates his profile*	
Comments: The test case is passed. Our system is working according to our need.			
■ *Passed* ☐ *Failed* ☐ *NotExecuted*			

Table 6.12: New Admin Addition Test Case

This test case details for New Admin Addition.

New Admin Addition			
Test Case ID:	*12*	**QA Test Engineer:**	*Usman Ghani*
Test case Version:	*1*	**Reviewed By:**	*Abdullah Sohail*
Test Date:	*01-04-2024*	**Use Case Reference(s):**	*Admin Addition Use Case (8)*
Revision History:	*None*		
Objective:	*To test whether the root admin can add new admin successfully.*		
Product/Ver/ Module:	*Listing Module of Website*		
Environment:	*Internet is connected, and the website is running*		
Assumptions:	*The one performing actions is root admin.*		
Pre-Requisite:	*The admin must be logged in.*		
Step No.	**Execution description**	**Procedure result**	
1 2	*Click "Add Admin" button, on the admin main page.* *After providing details, click "Add Admin" button*	*System redirects to the page requiring details of new admin.* *System notifies that new admin is added.*	
Comments: The test case is passed and root admin is successfully able to add new admins.			
■ *Passed* ☐ *Failed* ☐ *NotExecuted*			

6.3 Test Metric

Following is the test case matrix for the test cases which have been performed. It provides important insights related to test cases.

6.3.1 Functional Tests

Table 6.13: Functional Test Metric

This table summarizes the key metrics related to functional testing.

Metric	**Purpose**
Number of Test Cases	18
Number of Test Cases Passed	18
Number of Test Cases Failed	0
Test Case Defect Density	0
Test Case Effectiveness	0

Chapter 7 User Manual

Following section contains the user manual for our platform. Our platform will have two types of users one will be admin and other will be the end users.

7.1 Admin

Following are the guidelines for the admin.

7.1.1 Getting Started

Follow these steps to get started.

- Find the admin login page.
- Enter your email and password. After that click on login. It will redirect you to admin panel or admin home. If the password was wrong it will not redirect you to admin home.
- After logging in you will be on admin home. On your left you can see a side menu if on a bigger screen. Otherwise menu will be at top depending upon screen size.
- On admin homepage, admin can see system related data like CPU usage, memory usage etc.
- From side menu you can navigate admin home and can move to different pages.

7.1.2 Add or Delete Admin

To add/delete admin, follow the steps given below.

- Click on admin management button in side bar.
- Now you will be on a new page. Here you will see a list of all admin.
- Click on the delete button in front of an admin to delete that particular admin.
- At the top of the page you will see a form. Fill the information of new admin in that form. After filling the form click on the create button. As a result a new admin will be created.

7.1.3 Listing Management

Admin can delete a listing if it breaks the platform rules. Following are the steps.

- Click on listing button in side bar.
- Now you will be on a new page. Here you will see a list of all the listings.

- Click on the delete button in front of a listing to delete that particular list.
- List will be deleted.

7.1.4 Add or Delete Blog Post

To add/delete a blog admin can follow the steps given below.

- Click on blog button in side bar.
- Now you will be on a new page. Here you will see a list of all the blog posts.
- Click on the delete button in front of a blog post to delete that particular blog.
- At the top of the page you will see a section where you can write a new blog. After writing the blog click on the publish button. As a result a new blog will be published.

7.1.5 Admin Logout

To logout as an admin follow the steps given below.

- Click on logout button in side bar.
- Your session will end.
- You will be redirected to admin login page.

7.2 End User

Following are the guidelines for the end user.

7.2.1 Getting Started

To begin using Bidding Bazaar, follow these instructions:

- Make sure you have internet connected to your device.
- Make sure your device has latest version of any compatible browser.
- Open browser and go to our website.
- After loading you will be redirected to home page of Bidding Bazaar, where you can start exploring the platform.

7.2.2 Car Posting

To post a car for bidding of platform, the steps are given below:

- Click on ”Post a Car” button on home page.
- A page will be redirected to page requiring details and images of the car.
- After analyzing of the uploaded images and the car model by our image classifier, you will be given a range of price suggested by our price predictor.
- You can set the price within the suggested range.
- After clicking the post button, your car will be posted and you will be redirected to home page.

7.2.3 User Login With Email

To login with email follow the steps given below:

- Click on ”Login” button on home page.
- A page will be redirected to page requiring email and password.
- After providing details, click ”Login with Email” button.
- You will be logged in and redirected to home page.

7.2.4 User Login With Google

To login with Google account follow the steps given below:

- Click on ”Login” button on home page.
- A page will be redirected to login page.
- Click ”Login with Google” button.
- A pop-up window from Google will appear.
- Follow the instruction in this pop-up window.
- You will be logged in and redirected to home page.

7.2.5 Bidding Feature

To perform bidding on the listed cars follow the step given below

- After posting the car on the platform.
- The owner of the listing enables bidding by clicking the ’Start Bid’ button.
- A timers starts for the bidding.
- Multiple user can click on the bid button and add their bids.

- At the end of the bid the highest bidder wins.

7.2.6 Listing

To perform operations on the posted listing follow the steps below:

- After adding the listing, the user can perform Edit and Delete operations
- If the user is the owner he can delete the posted listing.
- User can edit the listing

7.2.7 Blogs

To post blogs on the platform follow the steps below:

- Go the blogs page from the nav at top.
- Click on Add Blog button.
- Write the blogs content and heading.
- Post the blog by clicking on Post button

7.2.8 User Profile

To perform operations on the User's profile:

- User clicks on the profile button at the top right corner.
- User is redirected to the profile page.
- User can view his profile.
- User can edit the profile by clicking the edit button

7.2.9 User Recommendations

Our platform offers personalized recommendations, we store the your interactions such as placing bids on the cars, and analyze this information to understand your preferences and match them with other users to suggest you cars that match your interest.

Chapter 8 Conclusion and Future Work

The used car marketplace is a big industry in the world. It needs efficiency, automation, and transparency. Our platform Bidding Bazaar, an online website to list and bid on cars online, offers to solve these issues with the use of AI. Our Platform helps the user to make quick decisions and sell their cars in less time and without stress.

Before the development, we did a literature review of similar platforms and analyzed them. We have mentioned our features and software requirements in detail in the report, which includes the requirements and use cases. After that, we mentioned our approach and methods we are going to use with the high and low level design of the whole system. The development work includes building a website where a user can register, sign In, list a car for bidding, bid on other cars, have his profile, an admin panel for managing bids and users and AI modules.

Our platform uses mainly two AI modules. Firstly, a module checks if the car posted by the user is a car or not. This reduces the overhead of the admin and listing of fake and prank things making our system fast. Secondly, When a car is being listed, our AI system predicts the car's price using XGBoost model. This AI model is trained on the local dataset from Pakistan. To scrape the data, we have built a web scraper using beautiful soup library of Python, that scrapes data from Pakwheels. This helps our users to make quick decisions, and not worry about setting up a perfect price.

For future work, we will be improving the user interface to enhance user experience. In addition to this, we will add more admin controls for the better management of the website. Enhancing the overall performance and user flow are also part of our future goals. We will be enhancing the platform itself making it better for user experience.

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